AMPHORA COATING IP / PORTFOLIO STATUS

Seventeen Declare It. Nine Can Prove It.

2026-08-04 — an adversarial audit of our own portfolio

17 modules declare Phase-0 in their own front-matter. The CLI verifier derives 9.

All 8 modules in the gap fail on one identical gate, for one identical reason. That is section 02, and it is the first thing in this report.

209 autonomous fires / **78** adversarial verdicts / **18** modules attacked

54 refuted / **22 survived** / **2** blocked on humans / **43** repair landings

Most of what follows is us breaking our own work. That is the point: an audit loop that always succeeds is generating paper, not information. Every number in this document is computed from a pulled evidence artifact, not typed. Where a claim is contested or unverifiable, it says so.

Master manifest SHA-256

f5623b3171d65de2dc805b740aef71ad6163554dcf9b5b394d84181b3ad6bf08

AUDIT-ONLY / NO PROMOTION AUTHORITY / SUPERSEDES 2026-08-03

01 / EXECUTIVE

The one-paragraph read

WHERE THE PORTFOLIO ACTUALLY STANDS

The loop has now fired **209** times and produced **78** adversarial verdicts across **18** modules — **54 refuted**, 22 survived, 2 blocked on a human. **43** repair-and-verify landings closed defects the loop had found, and 3 of those verifies turned round and *refuted the repair they were checking*. The single most important finding is not a science result at all: **17 modules declare `phase: PHASE_0` in their own front-matter, and the `amphora-cli` verifier derives 9**. The 8-module gap is not a disagreement about quality — every one of them fails on the same single gate, because a test harness overwrote the exact evidence bytes that each module's signed evaluation receipt pins by hash. The signatures are all still valid; what broke is the *binding* between a signature and the bytes it attests to. The root cause is fixed and locked by a new governance test; the original bytes are gone and cannot be recovered; the way back is a human dual-identity ceremony whose tooling is built, tested and waiting.



78 adversarial verdicts against our own claims, 18 modules

Why a report full of refutations is the good version of this document. An audit that cannot fail is not evidence of quality, it is evidence of a weak audit. This machinery can now return "your claim is wrong", and in this cycle it did so 54 times — including 3 times against *our own repairs*. The claims that survived are worth more than they were a week ago; the ones that did not are defects found before a customer, a partner or a regulator found them.

What founders should do next

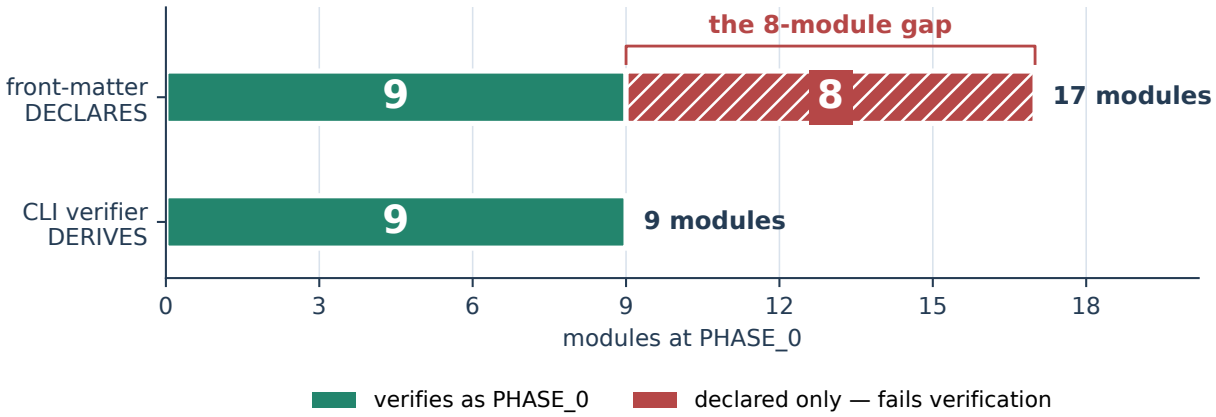
1. **Run the R-D07 re-evaluation ceremony.** It restores 8 of the 9 modules with a broken evaluation binding, and it is the only path that closes the 17-versus-9 gap. The tool is built, its refusal paths are tested, per-module dry-run plans exist, and no re-evaluation has been executed. It needs two human keys the loop deliberately does not hold. Sections 02 and 13.
2. **Decide R-D08 — the merge candidate.** `amphora-formulation-universe` has never been adversarially attacked, carries no science of its own, and its entire defensible claim is a strict subset of `amphora-formulation-arithmetic`'s. Merging would lose no defensible claim and would change every portfolio count. This is a portfolio-shape decision and the loop is forbidden from taking it. Section 13.
3. **Unblock the 6 data and rights asks.** R-D01 to R-D06 are each one human action — a rights confirmation, a counsel opinion, a data request, an author email. Between them they gate the coating-barrier science, the corrosion baseline, the rheology benchmark, the cure fixture, the design-rule sample size and the dielectric labels. Section 13.

02 / THE GAP

17 modules declare Phase-0. 9 derive it.

THE LEAD FINDING OF THIS CYCLE. IT IS A PROVENANCE DEFECT, NOT A SCIENCE DEFECT — AND IT IS WORSE THAN A SCIENCE DEFECT, BECAUSE IT IS LOAD-BEARING FOR EVERY PROMOTION CLAIM WE MAKE.

The measurement. Reading `phase:` out of every module's own `docs/mission.md` front-matter gives **17** modules at `PHASE_0`. Running the `amphora-cli` Phase-0 decision verifier over the same 29 mission-bearing modules derives **9**. The verifier's rule: *current phase is derived only from the latest valid module-local movement event with two distinct trusted signatures.* Report status is `EXECUTION_INCOMPLETE`; `authority_changes` is `False` — nothing was promoted or demoted to produce this number.



ONE failure mode: all 8 fail on exactly one gate, and it is the same gate

`EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt`

- cro-protocol-packager
- evidence-intake-validator
- known-answer-benchmarks
- molecule-design-gate
- optimizer-orchestrator
- redteam-audit
- report-packager
- source-ledger

The mechanism: two writers, one path, hash-pinned bytes

When `amphora-cli` runs `evaluate()` for a module it writes a benchmark result to `<module>/evidence/phase0/benchmark/result.v1.json`, hashes *the exact bytes it just wrote*, and seals that digest into the module's **immutable signed evaluation receipt** as `metrics.benchmark_result_sha256`. Later, `verify_evaluation_result()` re-reads the file on disk, re-hashes it, and compares. Independently of that, **27 per-module Phase-0 KPI integration tests** wrote their own fixtures *to the same path.* Any harness run therefore destroyed the evaluator-produced evidence and left behind bytes that hash to something else. The receipt is untouched and still verifies as a signature; the file it points at is no longer the file it was signed over.

What the verifier reports. Every module in the gap fails with **exactly one** gate, and it is the same gate in all 8 cases: `EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt.` No module in the gap fails any other gate. There is no second, hidden reason.

Module	Front-matter	Verifier	Sole failed gate
`cro-protocol-packager`	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt
`evidence-intake-validator`	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt
`known-answer-benchmarks`	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt
`molecule-design-gate`	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt

DSSE-signed, byte-stable evidence receipts on xbabe2 — 186 of 444 verify against a trust-anchored Phase-0 key; pre-loop receipts are tamper-evident, not anchored.

Module	Front-matter	Verifier	Sole failed gate
optimizer-orchestrator	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt
redteam-audit	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt
report-packager	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt
source-ledger	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt
molecule-property†	PRE-PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH: recorded benchmark is not bound to the evaluation receipt

† molecule-property is not part of the 17-vs-9 gap — its own front-matter already declares PRE_PHASE_0, so it was never counted as Phase-0. It carries the identical broken binding, which is why the affected population is 9 modules and the repairable population is 8.

02 / THE GAP

What is broken, what is not, and what cannot be undone

What is NOT affected

- **Every signature still verifies.** Evaluation receipts, independent review receipts and movement events all remain cryptographically valid. Nothing was forged, nothing was tampered with, no key was compromised. What broke is the *hash binding* between a signed receipt and the artifact it attests to.
- **No science changed.** Not one benchmark number, model, dataset or claim in the affected modules moved. This defect is invisible to every scientific KPI and would remain invisible for as long as nobody ran the verifier.
- **The 9 derived modules are unaffected.** candidate-contract, envelope-gates, formulation-arithmetic, formulation-universe, model-card-authority, process-qc-twin, run-provenance, solvency-score, tolerance-robustness — each has a signed evaluation, a signed independent review, a valid movement event and **zero** failed gates. Their Phase-0 status is derived, not declared, and it survives this finding intact.

Recovery was attempted. It is impossible.

The obvious repair is to put the original evaluator-produced bytes back. That was tried, methodically, and it failed:

1. **The method was validated first, on a healthy control.** Rather than assume a reconstruction procedure works, it was run against `amphora-solvency-score` — a module whose binding is intact — and confirmed byte-faithful there before being trusted anywhere else.
2. **It then failed for all 9 affected modules.** The reconstruction that reproduces a healthy module exactly does not reproduce any broken one.
3. **The bytes are absent from every history we hold.** A full sweep of git history and stashes, a content sweep of the entire worktree, and the 2026-07-22 xbabe1 archive all return harness-shaped bytes for all 9 modules. The clobber predates every snapshot in our possession.

Say this plainly. The original evidence is gone. Not misplaced, not quarantined — destroyed, by our own test harness, before any snapshot we still hold. No amount of autonomous work recovers it. The only honest path is to re-run the evaluation and re-sign it, which is a human act.

The root cause is fixed, and locked

- All **27** per-module Phase-0 KPI harness tests now round-trip through `CARGO_TARGET_TMPDIR` instead of writing the shared evidence path. The two writers no longer share a file.
- A new governance test, `shipped_benchmark_results_stay_bound_to_signed_evaluation_receipts`, fails if the bound set changes in *either* direction — it breaks the build if a binding is lost, and equally if one silently appears. It is not possible to quietly re-break or quietly re-fix this.
- The canonical suite is green with the fix in place: 62 suites / 378 tests.

The verifier was never relaxed to reach 9. The obvious way to make an unflattering number go away is to soften the checker. That did not happen, and it was audited independently rather than asserted: every guard function in the Phase-0 decision path was compared by SHA-256 and is **byte-identical** to its prior state. The number moved because the evidence moved, not because the test moved.

02 / THE GAP

The way back — and what could still be wrong with this section

WV-1 built the path, and did not walk it. The repair item WV-1 delivered a governed re-evaluation ceremony: `portfolio phase0 re-evaluate -module <M> -reason <text> -as-of <ts> -evaluation-key <external> -review <signed> [-dry-run]`. It refuses unless the module is on the known-unbound list *and* its shipped result is genuinely unbound by a live hash re-check (doubly enforced), refuses on a single signature, refuses a repository-local key, and archives the prior evaluation receipt, review, result and movement event under `evidence/phase0/superseded/<sha>/` with a supersession note. 6 refusal tests pass. Per-module dry-run plans exist. **No real re-evaluation has been executed.** It requires both the `PHASE0_EVALUATOR` and `PHASE0_INTERNAL_REVIEWER` external keys — dual identity is enforced, which is precisely why this loop cannot and must not run it.

8 are repairable. `molecule-property` is not.. The ceremony restores the 8 gap modules because each one still has a signed independent review and a valid movement event to reconcile against. `molecule-property` has **neither a review receipt nor a movement event**, so the verifier takes its live re-execution branch and fails against drifted portfolio state regardless of what is re-signed. It cannot be restored this way. Do not let a summary of this section round 8 up to 9.

What each repaired module actually gets

A successfully re-evaluated module lands at `PRE_PHASE_0` with `promotable=true`, and is then re-promoted with the existing `phase0 promote` command. The ceremony does not grant Phase-0; it restores the evidence that lets the normal promotion path be run honestly. Until a human runs it, **the correct number to quote externally is 9**, and any material citing 17 is wrong.

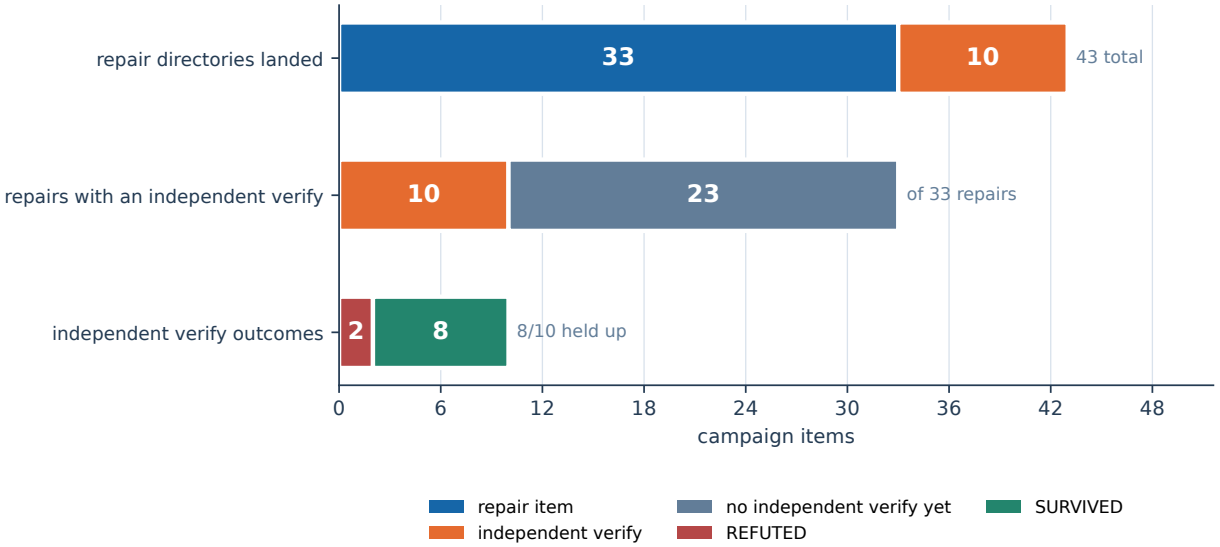
Red-team this section. Five things a sceptical reader should press on. **One:** the 9-versus-17 split is produced by *our* verifier over *our* front-matter; an external auditor should re-run `amphora-cli` themselves rather than take the table above on trust. **Two:** "the bytes are unrecoverable" is a negative claim, and negative claims are the hardest to prove — what was actually established is that they are absent from git history, from stashes, from a full worktree content sweep and from the 2026-07-22 archive. A copy could still exist somewhere nobody has thought to look. **Three:** the guard-function SHA comparison proves the guards were not *edited*; it does not prove they were correct to begin with. If the verifier was already too lenient, this finding under-reports the problem rather than over-reporting it. **Four:** the campaign item that carries this work states the honest count as "9, not 16", while front-matter across the tree declares 17 — a one-module discrepancy between the in-code assertion and the checked-in front-matter that is not yet reconciled, and that this report resolves in favour of the front-matter because the front-matter is what a reader sees. **Five:** nothing here has been reviewed outside this portfolio, and the ceremony that fixes it has never been executed even once, so its refusal paths are tested but its success path is not.

03 / REPAIRS

43 landings — and the two that refuted themselves

A REFUTATION THAT NEVER GETS REPAIRED IS A COMPLAINT. A REPAIR THAT NEVER GETS INDEPENDENTLY VERIFIED IS A HOPE. THIS SECTION REPORTS BOTH HALVES.

What landed. **43** repair-and-verify landings across **18** modules: 30 repair rows carrying 26 distinct repair items, and 13 independent verify rows. Of the 26 distinct repair items, **11** have been checked by a separately-authored verify and **15** have not yet been. An unverified repair is not a closed defect; it is a claim about a defect.



Module	N	Repairs landed	Independent verifies
molecule-property	5	B3, R-A09, R-A11, R-N02	V-N02a
redteam-audit	5	B2, PHASE-D, R-A00, WV-1	V1-RESOLVE
molecule-role-ranker	4	R-A12, R-N02	V-A12a, V-N02b
public-corrosion-benchmark	4	R-A08, R-N03	V-A08, V-N03
sourceability-ehs	4	R-A04, R-A12, R-N01	V-A12b
molecule-generator	3	R-A06, R-A06v2	V-A06
molecule-novelty	3	R-A07, R-A07v2	V-A07
solvency-score	3	R-A10, R-B03	V1-SOLV
known-answer-benchmarks	2	R-A05	V-A05
rdkit-ffi	2	R-N02	V-N02c
assurance-capabilities	1	R-B01	—
barrier-corrosion-score	1	R-B02	—
cure-potlife-gate	1	R-A02	—
design-rule-score	1	R-A11	—
ingredient-library	1	R-A01	—
ip-fto-guard	1	R-A13	—
model-card-authority	1	—	V1-KPI-MCA
viscosity-score	1	R-A03	—

03 / REPAIRS

The credibility exhibit

13 VERIFY ITEMS RAN AGAINST LANDED REPAIRS. 10 CONFIRMED THE REPAIR. 3 REFUTED IT.

Verification that never refutes is not verification. The point of pairing every repair with a separately-authored verify is that the verify has to be able to *lose*. Two did, on consecutive fires, and both times the failure was one the repair's own self-check could not have caught — because the verify escalated past the repair's chosen frame instead of re-running it.

V-A06 — the arithmetic was right and the label was wrong

Against `amphora-molecule-generator`'s R-A06 repair. The verify first *reproduced the repair exactly*: a fresh RDKit re-derivation of `amine_sites` and `active_nh` straight from raw SMILES matched, and the Clopper–Pearson interval and family IntDiv1 reproduced to four decimal places. It then escalated past the self-check by testing the repair's corrected "29 of 38 difunctional" headline against the **active-hydrogen (AHEW) functionality convention this portfolio's own receipts already use** — under which all 29 are `active_nh=4` primary diamines, including MDA/DDM, the textbook *tetra*functional curing agent. Under the convention we ourselves apply elsewhere, **0 of 38** accepted proposals are difunctional. A correct calculation of the wrong quantity. Superseded by `R-A06v2`.

V-A07 — the fix was right and reached one place out of four

Against `amphora-molecule-novelty`'s R-A07 repair, which corrected a genuine structural defect: the signed IPDA candidate SMILES canonicalised to $C_9H_{20}N_2$, a demethylated non-molecule, not real isophorone diamine ($C_{10}H_{22}N_2$, CAS 2855-13-2). The repair propagated that correction into a *Morgan-only side table*. The verify propagated it into **all four** fingerprint families — `maccs_166`, `atompair_2048`, `toptorsion_2048` as well as Morgan — and diffed against the shipped table. One cell flips that the receipt does not disclose: `atompair_2048` / IPDA / α=0.01 moves NOT-NOVEL → NOVEL. The repair's claim to have corrected "the" table was false; it had corrected a quarter of it. Superseded by `R-A07v2`.

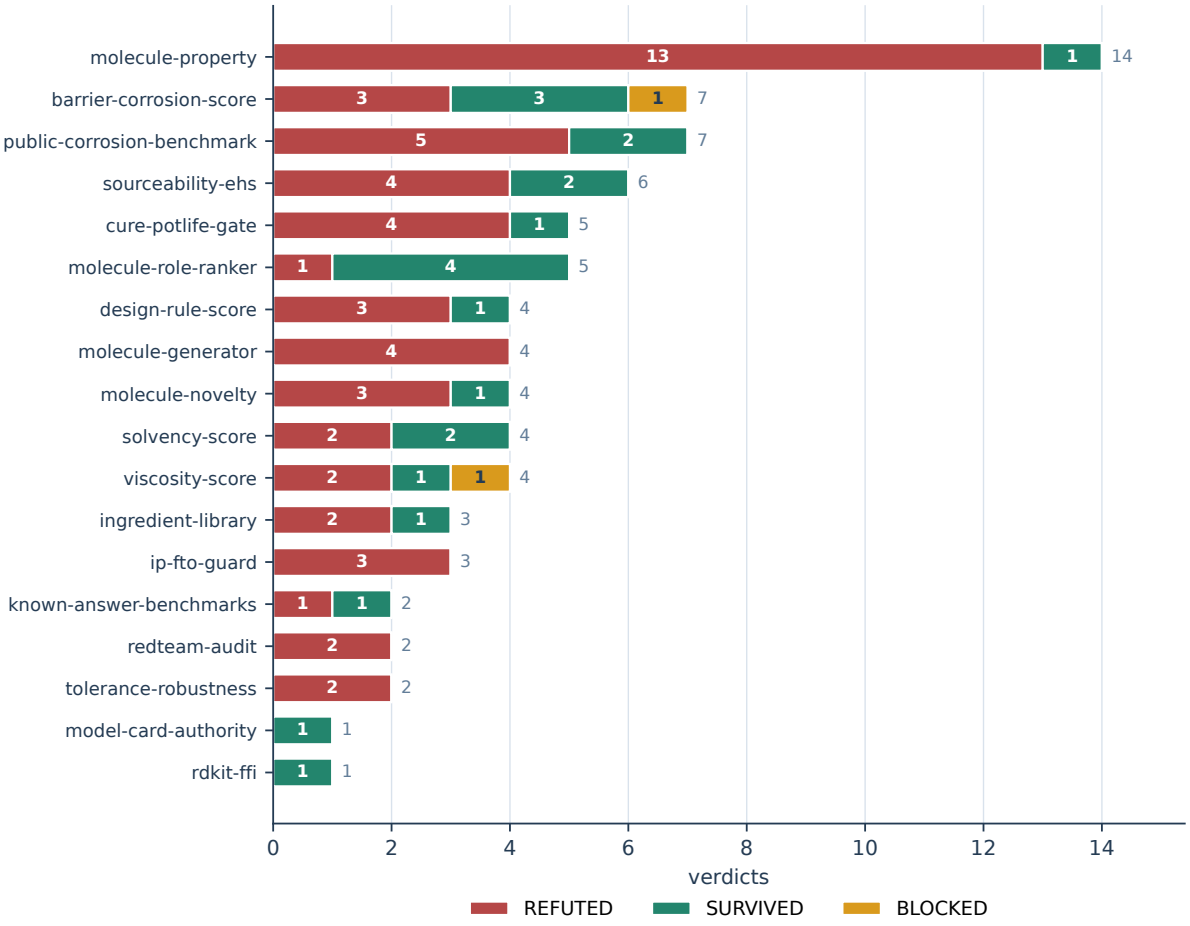
The third refusal. A third verify, `V1-RESOLVE`, refuted a portfolio-wide claim rather than a single repair: the assertion that every module with a refuted verdict carries a resolving retraction index. It resolved all 80 rows across 16 module indexes row by row — exists, DSSE-verifies, and actually covers the claim it says it covers — and found the assertion did not hold. That is the same failure mode as V-A06 and V-A07: the original check tested presence, the verify tested sufficiency.

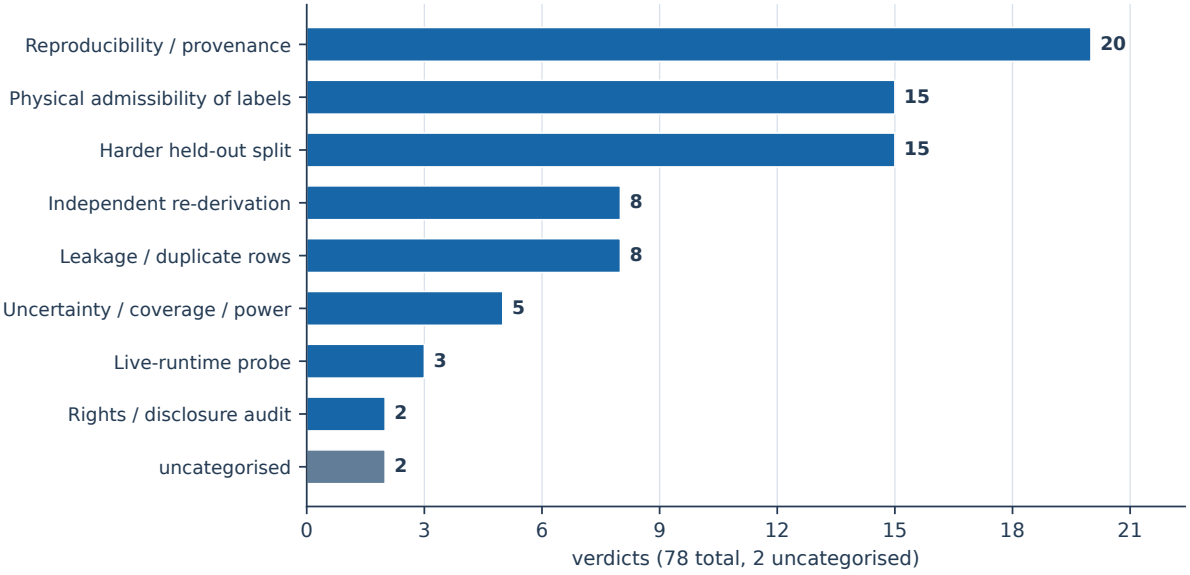
Red-team this section. Three caveats. **One:** the verifies are authored by the same loop as the repairs. They are independent in *implementation* — fresh harness, fresh derivation, novel probes never in the repair's probe set — but not in *interest*. **Two:** 15 of the 26 distinct repair items still carry no independent verify at all, so the 3-in-13 refutation rate above is measured only over the repairs we chose to re-check. If that rate holds on the remainder, roughly 3 further landed repairs are wrong. **Three:** a v2 repair is not evidence of correctness either — `R-A06v2` and `R-A07v2` have themselves not been independently verified.

04 / SCOREBOARD

Where the attacks landed

78 VERDICTS ACROSS 18 MODULES. 12 MISSION-BEARING MODULES HAVE NEVER BEEN ATTACKED AT ALL — THAT IS A GAP, NOT A PASS.





What the attacks were made of. 77 fires carry a self-declared attack type. Grouped: data / label integrity (22), other / one-off (22), generalisation & split hygiene (9), completeness sweeps (7), rights & provenance (6), statistical admissibility (4), verify-of-repair (4), baseline strength (3). The distribution is the tell — the loop is overwhelmingly finding *data and provenance* defects rather than modelling defects, because that is where the portfolio is weakest.

04 / SCOREBOARD

Verdicts by module, and the untested remainder

Module	All	Ref	Surv	Blk	Latest verdict
molecule-property	14	13	1	0	08-04 01:06
barrier-corrosion-score	7	3	3	1	08-03 11:10
public-corrosion-benchmark	7	5	2	0	08-04 08:07
sourceability-ehs	6	4	2	0	08-04 10:07
cure-potlife-gate	5	4	1	0	08-03 12:06
molecule-role-ranker	5	1	4	0	08-04 09:22
design-rule-score	4	3	1	0	08-02 21:05
molecule-generator	4	4	0	0	08-04 06:07
molecule-novelty	4	3	1	0	08-04 07:13
solvency-score	4	2	2	0	08-04 12:43
viscosity-score	4	2	1	1	08-03 06:13
ingredient-library	3	2	1	0	08-03 14:06
ip-fto-guard	3	3	0	0	08-03 15:08
known-answer-benchmarks	2	1	1	0	08-04 05:08
redteam-audit	2	2	0	0	08-04 11:13
tolerance-robustness	2	2	0	0	08-03 10:05
model-card-authority	1	0	1	0	08-04 13:09
rdkit-ffi	1	0	1	0	08-04 03:19
Total	**78**	**54**	**22**	**2**	

The 12 never attacked — and what is queued against them. candidate-contract [T-01], cro-protocol-packager [T-02], envelope-gates [T-03], evidence-intake-validator [T-04], formulation-arithmetic [T-05], formulation-universe [T-06], molecule-design-gate [T-08], optimizer-orchestrator [T-09], process-qc-twin [T-10], report-packager [T-12], run-provenance [T-13], source-ledger [T-14]. These are not generic "find something" rotations. Each is a campaign item (T-01 – T-14) with a module-specific probe written against that module's own declared claim, so a survival will mean something. Two further items in that series, T-07 and T-11, target modules that have since been attacked and remain queued for a second pass.

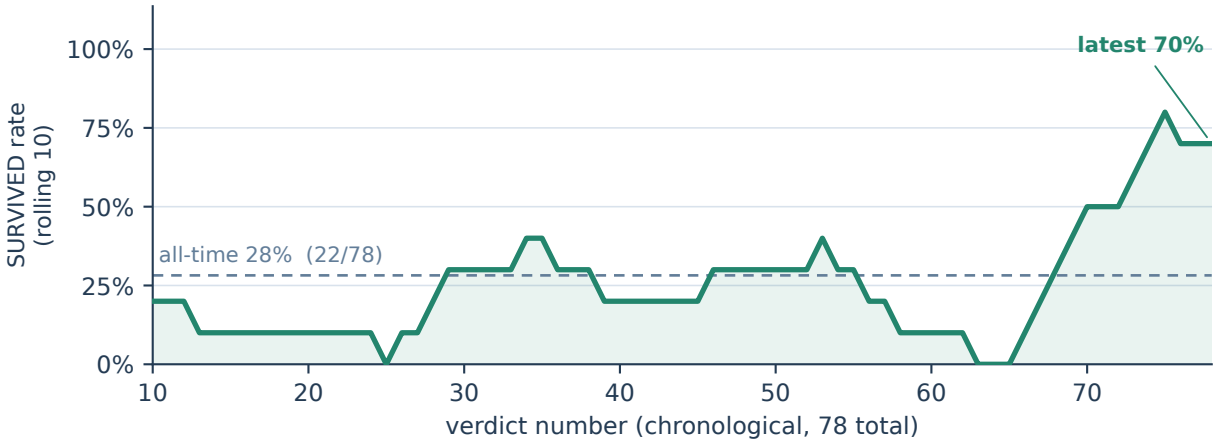
Red-team this section. Attack counts are not quality scores. amphora-molecule-property carries 14 verdicts largely because it is the most instrumented module in the portfolio — density of attack follows density of existing evidence, so the best-documented module looks like the worst one. The inverse error is more dangerous: the 12 modules with no bar at all are not clean, they are unexamined, and several of them sit inside the Phase-0 gap in section 02. Treat an unattacked module carrying a composite score as a liability, not an asset.

05 / LOOP OPERATIONS

The machine that produced this report

CONDENSED. THE FULL TREATMENT — PER-FIRE ACCOUNTING, EVERY INFRASTRUCTURE MODULE, THE CAMPAIGN STATE MACHINE — IS IN LOOP-OPERATIONS-2026-08-04.PDF, SHIPPED ALONGSIDE THIS DOCUMENT.

Shape of the run. 209 fires between 2026-07-26 and 2026-08-04, median duration 8 min 55 s. Modes: (pre-campaign) 127, CAMPAIGN 43, ROTATION 39. Outcomes: 45 CLAIM_REFUTED, 18 CLAIM_SURVIVED, 2 BLOCKED_RIGHTS, 18 UNKNOWN, 11 SATURATED_SKIP, and 115 legacy SIGNED_RECEIPT from before the outcome vocabulary existed. 196 receipts were added across the run.



Survival convergence

Across the 65 fires that returned a graded verdict, the survival share is **28%**. Split at the midpoint of the graded sequence it moves from **19%** in the first half to **36%** in the second. Read that carefully in both directions. The optimistic reading is that the loop ran out of easy defects: the early fires attacked flagship claims that had never been probed, and the later ones increasingly attacked *repairs*, which are newer, narrower and built in response to a known failure. The pessimistic reading is that a loop grading its own attacks will drift toward attacks it can survive. Section 03 is the control for that reading — 3 of the late fires refuted our own repairs, which is not what attack-softening looks like.

The signing-violation fix

The launcher's signing checker raised **110** violation alerts over the run, peaking at **59** on 2026-08-02. Almost all were false positives from a checker probing a field name that does not exist in the schema it was reading — forensically established, then fixed. On 2026-08-04 the count is **0**.

Day	Fires	Signing violations
2026-07-26	1	0
2026-07-27	23	1
2026-07-28	24	1
2026-07-29	23	1
2026-07-30	23	0
2026-07-31	25	1
2026-08-01	25	2
2026-08-02	25	59
2026-08-03	24	45
2026-08-04	16	0

05 / LOOP OPERATIONS

Wasted fires, and a defect found in our own reporting data

Wasted-fire accounting. Not every fire produced information. **11** returned SATURATED_SKIP — the campaign item had exhausted its fire budget and the loop correctly declined to burn another. **18** returned UNKNOWN, meaning the launcher's independent ssh re-check *rejected* a verdict the agent claimed. **17** fires produced no verified new signed artifact at all, and **4** claimed a verdict that could not be found in the ledger under that fire's own id. Taken together that is on the order of **46** of 209 fires that moved nothing — roughly 22%. We report it rather than netting it out, because a loop that hides its own no-ops cannot be cost-managed.

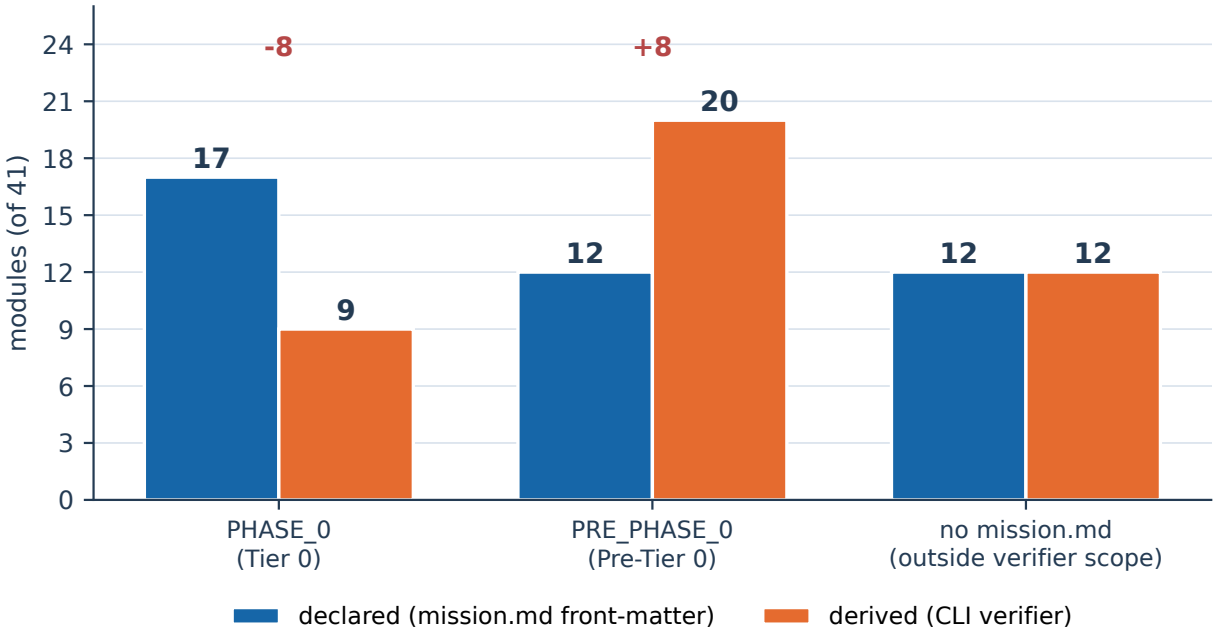
Found and fixed while building this report: corrupted timestamps in the verdict ledger. Parsing `logs/falsification-ledger.jsonl` for this report surfaced **2** rows whose `ts` field carries a stray leading backslash — `/2026-08-03T15:08:37+00:00` rather than a bare ISO-8601 stamp. Affected modules: ip-fto-guard, sourceability-ehs. Any consumer doing a strict `fromisoformat` parse would either crash or silently drop those verdicts, and a lexical sort would file them at the wrong end of the sequence. The render pipeline for this report now strips and counts them, and this box is the disclosure. It is a small defect and it is exactly the class of thing that makes a ledger untrustworthy: the two affected rows are real verdicts — one against `amphora-ip-fto-guard`, one the `V-A12b` verify — that a naive reader would have lost. Full remediation status is in `loop-operations-2026-08-04.pdf`.

How a verdict is verified. Three checks, all run by the launcher over ssh rather than by the agent that claims the verdict. **Content-hash attribution** — the artifact must be new or changed by SHA-256 against a pre-fire snapshot and must be one the agent explicitly declared, so touching a file or inheriting a concurrent agent's work cannot be credited. **Real Ed25519 DSSE** — the receipt must verify against the pinned Phase-0 key over the DSSE pre-authentication encoding, not merely look like a signature object. **Fire-id ledger binding** — the ledger must contain a line carrying this specific fire's id. The 18 UNKNOWN outcomes above are what happens when a claimed verdict fails one of the three.

Red-team this section. The verification chain constrains *provenance*, not *scientific quality*. It proves a verdict arrived with genuinely new, correctly signed evidence produced by the fire that claimed it. It cannot prove the attack was well designed, and a weak attack reported as CLAIM_SURVIVED passes all three checks cleanly. The median fire runs 8 min 55 s; anything materially faster is worth opening for exactly this reason. Full per-fire detail, including the runtime distribution and the campaign state machine, is in `loop-operations-2026-08-04.pdf`.

06 / SYSTEM

Portfolio state: declared, derived, and counted



Count the crates carefully. The tree holds **41** `amphora-*` crates. Only **29** carry a `docs/mission.md`, and only those 29 make any scientific or contractual claim — the other 12 are runtime plumbing with nothing to falsify. Every phase number in this report is computed over the 29, which is the same roster the CLI verifier uses. `amphora-solver-lab` **is new this cycle and has no** `mission.md` **yet**, so it appears in the crate count and in section 10 but in no phase or tier count. Quoting "41 modules" externally would be an overstatement of the claim surface by 12.

Declared tiers versus derived phase. By front-matter: 17 Tier 0, 12 Pre-Tier 0, 0 Tier 1, 12 infrastructure crates with no mission file. Within Tier 0, 4 carry an actual scientific claim (`LOCKED_INTERNAL`) and 13 are contract and plumbing modules that assert no science by design. By the verifier: **9 at PHASE_0 and 20 at PRE_PHASE_0**. Where those two disagree, section 02 explains why, and the derived number is the one to use.

Evidence, not status. **74** Phase-0 KPI files across **26** modules; **444** receipt files in the tree; the founder-signed master promotion manifest remains at **66** receipts across 11 modules. **Zero modules were promoted this cycle.** The loop produces AUDIT-ONLY evidence and promotion requires a human dual-identity ceremony it deliberately cannot perform — and this cycle the honest movement was in the other direction.

Red-team this section. A reader could fairly ask why a week of intensive work produced no tier movement and a *lower* derived Phase-0 count than the declared one. That is the correct outcome, and it is worth being blunt about why: the work did not fail to find promotions, it found reasons not to promote. The composite scores on the tier chart are computed from evidence *density* — receipt counts, baseline presence, split methodology, uncertainty quantification — and say nothing about whether a claim survives contact with an adversary. 12 mission-bearing modules carry a score and have never been attacked. Read the chart as "how much evidence exists", never as "how right this is".

07 / LIFT

Where we beat best known practice — and where the baseline is a straw man

ONLY COMPARISONS WITH A GENUINELY DEFENSIBLE PUBLISHED BASELINE APPEAR IN THE FIRST TABLE. EVERY VALUE IS QUOTED FROM A SIGNED RECEIPT. NOTHING IS DROPPED FOR BEING UNFLATTERING.

Property	Ours	Published baseline	Base	Δ	Split / n
Hansen χ	**MAE 0.656** χ	Hildebrand–Scott 1950 single-δ	1.730	−62.1%	a-priori, zero fitted params, n=8 published pairs
BigSolDB solubility	**RMSE 0.877 log**	best naive (solvent-mean)	1.169	−25.0%	leave-solute-out, 1,372 solute groups, n=94,532
Density	**MAE 0.0566 g/cm^3**	Van Krevelen 1990 Ch.4, 0 params	0.1327	−57.3%	random 60/20/20 — *not* group-safe, n=1,077
Glass transition T_g	**MAE 51.24 K**	Van Krevelen Ridge Y_{gi}	70.42	−27.2%	LOSO-CV, 212 scaffold folds, n=441
Refractive index	**MAE 0.0304**	Lorentz–Lorenz, real MD density	0.0377	−19.3%	source-stratified 60/20/20, n=1,223
Melt viscosity‡	**R^2 0.4536 (ML)**	Fox–Flory–Whitten + Arrhenius	0.5972	physics **wins**	polymer-safe by SMILES, only n=13 test chemistries
Cure kinetics E_a	**57.261 kJ/mol**	Kissinger 1957 on CC-BY DSC	57.3 pub.	err 0.039	3 systems, 3 DOIs, cross-lab

Two results worth reading twice. On **melt viscosity** classical polymer physics *beats* our ML by +0.1436 R^2. On **cure kinetics**, three independent published isoconversional estimators all recover activation energies that produce *lower* pot-life error than the value we currently cite (9.3% vs 13.3% MAPE). Both are cases where the hardest available baseline won, and both are reported here rather than buried.

‡ **The melt-viscosity row is contested, and stays in with a flag.** The physics-beats-ML *arithmetic* is not disputed — an RMSE/R^2 self-consistency control across all five baselines confirms the numbers. What was refuted (2026-08-03) is the KPI's standing as a documented finding: an exhaustive scan of the KPI JSON finds **no bootstrap, CI, cluster, jackknife or standard-error token anywhere in it**, while its own sibling conformal KPI on the identical seed-0 41-train/13-test partition explicitly discloses a small-n conditional-coverage collapse. Thirteen independent test polymers is far below the $G \geq \sim 30$ cluster-count adequacy threshold (Cameron–Gelbach–Miller 2008; Cameron–Miller 2015). The honest statement is "physics beat ML by +0.1436 R^2 on 13 chemistries, with no uncertainty quantification of any kind" — which is still an interesting result and is not a benchmark.

COMPARISONS WHOSE BASELINE IS NOT THE HARDEST DEFENSIBLE ALTERNATIVE — 8 FLAGGED

Comparison	Why the Δ is not yet defensible
Flory–Huggins / Hansen "ML beats theory"	**Was void, now re-run fairly.** The original comparator collapsed onto constant-mean (fitted slope $\sim 10^{-35}$) with δ values exceeding the admissible envelope by up to 16 orders of magnitude, so the comparison was meaningless in both directions. `R-B03` rebuilt the physics arm as a coverage-gated Fedors/Van-Krevelen instrument and `V1-SOLV` (CLAIM_SURVIVED, 2026-08-04) re-ran the comparison on it — identical scaffold-safe 60/20/20 seed-0 split, identical HistGBM arm, test n=5,261, two-replay byte-identical, physics R^2 scored only over instrument-*returned* molecules with abstentions excluded and counted (never zero-filled), coverage 85.5% Fedors / 75.1% Hansen. It ran fairly and stayed non-degenerate in **5 of 6 arms**. The per-arm deltas live in the V1-SOLV receipt and are deliberately not restated here: the pulled summary is truncated, and one arm did collapse.
Pot-life Arrhenius lift 0.805×	Baseline is *no temperature correction at all.* A chemistry-blind Q10=2 heuristic matches it (MAPE 0.1527 vs 0.1473) and **beats** it at 30 °C and 40 °C, so the published activation energies are not demonstrably load-bearing. Envelope now disclosed by `R-A02`.
Viscosity Arrhenius lift +64 pp	Baseline is constant viscosity at 25 °C. No named correlation (Andrade, VFT, Vogel) is benchmarked.
Barrier diffusivity R^2 +0.269 vs −0.061	The physics arm is a *simplified surrogate*; the KPI itself concedes that full Vrentas–Duda 1977 is Phase-1 scope.
NIST corrosion +0.047 log10	Null-median baseline, and **refuted twice**: Ridge beats the null on 17/30 folds (p=0.585, bootstrap CI crosses zero), and an independent re-implementation with per-fold alpha tuning and two fresh seeds did no better (16/30, Wilcoxon p=0.262). Superseded by `R-A08` + `R-N03`; the corrected position is *not significant in-domain, negative under extrapolation*, and that correction survived `V-A08` and `V-N03`.
Design-rule leak-free ΔR^2	Paired bootstrap CI [−0.022, +0.568] crosses zero at n=58 unique observations. Binding constraint is sample size, not model form — which is what `R-D05` asks a human to fix.
Tg vs Van Krevelen (PolyUniverse)	The rescaled VK arm scores R^2 −0.042 — statistically indistinguishable from constant-mean. "ML beats VK by +0.87 R^2" is really "ML beats a constant."
molecule-property NULL arms	Every "beats NULL" gate in that module compares against a median. Published group-contribution tables exist and are only partly used.

Red-team this section. The first table is the strongest material in the portfolio and it still carries three structural caveats. **Split methodology is not uniform**: density and refractive index use a plain random 60/20/20 split, and every RadonPy property fails a chemically-coherent held-out split when one is imposed — those two rows describe within-chemotype performance, not generalisation. **No time-safe split exists anywhere in the portfolio**; the one temporally ordered dataset turned out to be cumulative snapshots that were 64% leaked. **Properties with no defensible baseline at all** — dielectric, coating-barrier efficacy, pigment crowding, thermal conductivity — are absent from both tables by construction, which is itself the finding. And the one row that moved in our favour this cycle, the solvency ML-vs-physics re-run, moved from *void* to *fairly run*, not from *losing* to *winning*.

08 / DEEP DIVE · AMPHORA-MOLECULE-PROPERTY

molecule-property

COMPOSITE 73.5 / 21 KPI RECEIPTS / 97 RECEIPT FILES, 11 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 14 verdicts — **13 refuted**, **1 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 1 failed gate: EVALUATION_RESULT_MISMATCH. Repairs landed: B3, R-A09, R-A11, R-N02, V-N02a.

Strongest defensible claim today

The corrected active-N-H amine classification rule (R-N02), which V-N02a re-derived from live `n_active_amine_nh()`, re-ran across all 26 asserted records with zero regressions, and then escalated with three amidine-family probes and an aromatic over-firing boundary probe never in the repair's set. It is a structural rule, not a predictive model — and that is the point. Every *predictive* headline in this module has been refuted or narrowed: density MAE 0.0566 g/cm^3 does not transfer out-of-corpus, the dielectric arm has no competent model on the physically-valid subset, and the multi-target RadonPy R^2 values rest on a random rather than chemically-coherent split.

When	Verdict	Attack
08-01 05:13	REFUTED	C7: cross-corpus external-validity (OpenPoly Tg RF <-> Polyuniverse Epoxy MD Tg, both directions) + leave-polymer-class-out proxy for density (RadonPy PI1070, no...
08-01 16:25	REFUTED	C7 fire 2: density escalation past leave-polymer-class-out (1.80x SURVIVED). Tried leave-one-Butina-cluster-out (dead end, 0.95x, documented). Primary:...
08-01 17:04	REFUTED	C7 fire 3 of 3: reconciliation + DSSE signature-format fix for fires 1/2 (both had correct science but bare-string .signature). Re-verified fire1/fire2 diagnostics...
08-01 18:18	REFUTED	C8 fire 1: built the consolidated 4-property {physics,ML,stack} comparison receipt (density: assembled from existing; Tg: new stack arm built on OpenPoly;...
08-01 19:04	REFUTED	C8 fire 2: escalated fire-1 dielectric data-integrity finding into a zero-contamination pipeline – RF/stack retrained from scratch using ONLY the 113/1075=10.5%...
08-01 20:02	REFUTED	C8 fire 3 of 4: reconciliation, not a new experiment. Consolidated fire1 (density/Tg/RI triplets + dielectric label-integrity finding) and fire2...
08-02 01:07	REFUTED	W2 fire 1: generalized the C8 physical-admissibility audit (eps_static>=n^2 on RadonPy PI1070) to 4 in-tree corpora – OpenPoly Tg/Tm ordering (Boyer-Beaman),...
08-02 02:03	REFUTED	W2 fire 2: re-sign of fire-1 4-corpus physical-admissibility audit (OpenPoly Tg/Tm, BigSolDB solubility+density, RadonPy polyomics_small_molecules...
08-02 03:04	REFUTED	W2 fire 3: root-caused fire1/fire2 rejection (signing_check=OK but verified_new_signed=0) as a KEY-ROLE bug, not a key-material bug – fire2s trust-anchored...
08-02 04:05	REFUTED	W3: propagate C8 dielectric-label-inadmissibility finding to every consuming KPI/report surface; write superseding retraction receipt + machine-readable retraction...
08-02 14:05	REFUTED	W7: hierarchical average-linkage macro-bisection (identical method to C7 fire 2 density attack) on RadonPy PI1070 4-property-complete subset (n=1068), applied to...
08-03 02:09	REFUTED	W8: physical-admissibility audit of the ridge-fit KPIs own stored fitted-vs-published Y_gi coefficient table. 2/24 fitted Y_gi are sign-reversed vs their published...
08-03 18:08	REFUTED	guanidine-conjugation internal-consistency probe: extracted shipped n_active_amine_nh() verbatim, ran unmodified on guanidine/dicyandiamide(DICY)/biguanide – all...
08-04 01:06	SURVIVED	V-N02a independent verify: fresh extraction of live n_active_amine_nh(), re-ran original 3 guanidine/DICY/biguanide probes + full 26-record zero-regression check...

Red-team. Fourteen verdicts is the most of any module and thirteen are refutations, but do not read that as "worst module" — it is the most instrumented, so it offers the most to attack. The real residual risk is different: this module sits in the Phase-0 gap of section 02 and is the one affected module the R-D07 ceremony *cannot* restore, because it has no review receipt and no movement event. Separately, its surviving property claims are all within-corpus; cross-corpus transfer was tested and failed, so the honest applicability domain is narrower than the KPI surface implies.

08 / DEEP DIVE · AMPHORA-BARRIER-CORROSION-SCORE

barrier-corrosion-score

COMPOSITE 58.1 / 2 KPI RECEIPTS / 26 RECEIPT FILES, 11 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 7 verdicts — **3 refuted**, **3 survived**, 1 blocked. Verifier state: PRE-PHASE-0, 2 failed gates: SCIENTIFIC_BENCHMARK_NOT_EXECUTED; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-B02.

Strongest defensible claim today

Within-source conformal coverage on Polyverse diffusivity (0.9277 against a 0.90 target, with the worst applicability-domain quintile also above floor), which survived a temperature-regime extrapolation shift across ten scaffold-partition seeds. The corrosion-rate QSPR also survived a metallurgical-family re-partition, but only just — R^2 fell from 0.316 to 0.132 with one fold going negative, and it survived solely because the pre-registered bar was $R^2 > 0$.

When	Verdict	Attack
07-31 22:06	BLOCKED	rights-package-assembly: hash-bound RIGHTS-REVIEW-REQUEST.md + consolidated data-acquisition-queue entry for USBR ST-19168 (catalog 4640/item 11562, sha 9e34fccc..., 10 coating systems) referencing the two already-cleared public-corrosion-benchmark USBR asks (17 U.S.C. 105); xlsx untouched, not opened/parsed; claim neither...
07-31 23:47	SURVIVED	pre-registered+hash-locked the frozen ST-19168 analysis pipeline (exact/MC permutation primary test, fixed-sequence bootstrap superiority gates) and stress-tested it on pinned-seed synthetic shadow data (10 systems x EIS|Z|+rust-creep, same shape as the still-quarantined real file, zero real content used): K=15 null-arm draws...
08-01 00:08	REFUTED	genuine 5-query literature sweep of MDPI Coatings/Materials/Polymers, ACS Omega/AuthorChoice, RSC OA, Sci Reports/Nat Comms (2020-2026) for a third strictly in-class open table-read rights-clean epoxy-amine/polyamide EIS ladder; found Alam et al. Materials 15(7):2562 (2022) DOI 10.3390/ma15072562 PMCID PMC8999854, CC-BY-4.0...
08-02 00:05	REFUTED	environment-domain-shift conformal: re-partition SAME held-out rows by marine/chloride vs not (instead of material-bucket parity); reverse direction (cal=marine, test=nonmarine) shows 0.90-target coverage=0.876, one-sided binomial p=3.1e-5
08-02 00:26	REFUTED	first-ever real-data execution of the never-run locked KPI test: adapted the frozen st19168-preregistered-v1 protocol (primitives+constants unchanged) to the 3 CC-BY literature series (asperti2024 n=4, zhu2022 n=5, alam2022 E-series n=3) unlocked by C4, within-study only, no cross-study pooling, honoring each source...
08-02 19:11	SURVIVED	temperature-regime extrapolation shift within ncomm(2023): calibrate on T<=308K, test on held-out T>308K (293-413K full range), paired vs matched-T control across 10 scaffold-partition seeds, one-sided paired t-test on shift-minus-control coverage; also isolates the AD-flagged-in-domain subset since the receipts own...
08-03 11:10	SURVIVED	family-safe re-partition: reran corr_rate_qspr.py feature engineering/model/seed verbatim but grouped GroupKFold by metallurgical family (ASM Vol.3/Sedriks1996/Uhlig-Revie2008, 17 families, 94.58pct coverage) instead of raw material string, forcing near-duplicate alloy aliases (316/316L/316LN/316Ti;...

Red-team. The module's locked KPI ranks by `metrics.barrier_index`, which the live implementation did not compute at all; R-B02 landed against that and has *not* yet been independently verified. The three available literature series are individually too small to reach significance regardless, and the one asset that would change that — USBR ST-19168 — is blocked on a human rights confirmation (R-D01). Until then this module's science is gated on paperwork, not on modelling.

08 / DEEP DIVE · AMPHORA-PUBLIC-CORROSION-BENCHMARK

public-corrosion-benchmark

COMPOSITE 41.2 / 3 KPI RECEIPTS / 15 RECEIPT FILES, 12 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 7 verdicts — **5 refuted**, **2 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 1 failed gate: EVALUATION_RESULT_MISMATCH. Repairs landed: R-A08, R-N03, V-A08, V-N03.

Strongest defensible claim today

A negative result, correctly stated: the NIST CORR-DATA marine-steel Ridge baseline is *not* significant in-domain and is negative under extrapolation. That correction (R-A08, R-N03) is the most heavily verified position in the portfolio — V-A08 re-implemented it from scratch with per-fold alpha tuning and two fresh bootstrap seeds and got no better (16/30 folds, Wilcoxon p=0.262), and V-N03 independently re-ran the committed generator two-replay byte-identical.

When	Verdict	Attack
08-01 23:04	REFUTED	completeness audit of KPI surface vs evidence/audit/ tree: checked whether the 4 signed coating-side audit verdicts (Brasher-Kingsbury H0_REFUTED 6/6 + T0 data-integrity erratum, EIS discrimination-power n~90 required, cross-modality sign-confound, multi-instrument W=0.492 weak-or-null) were indexed anywhere under...
08-02 05:05	REFUTED	exhaustive tree sweep (40 amphora-* dirs, 29 with an evidence tree): decode every DSSE-signed json under evidence/audit/+benchmarks/, extract verdict fields, test path/base-name/sha256 substring membership in that module own evidence/phase0/kpis/ blob, tally unindexed and negative-unindexed per module
08-02 16:10	REFUTED	reproducibility + fold-level significance audit: independently re-derived n=588/30-doc population from raw hash-pinned NIST CORR-DATA CSV (exact match); found the generating script (/tmp/corr_baseline.py per receipt_meta) was never committed and no longer exists, so the pinned Ridge MAE=0.6331 is not independently...
08-03 04:04	REFUTED	staleness/self-consistency audit: real-Ed25519-verified all 6 signed evidence/audit/ receipts against the pinned Phase-0 evaluator key, set-diffed against the consolidated KPIs own audit_verdicts_index (claims 4), found 2 unindexed (kpi-surface-completeness-v1, nist-baseline-fold-significance-v1); the latter directly targets...
08-03 20:05	REFUTED	leave-one-material-group-out (LOMO-CV) extrapolation probe: a genuinely new, group-safe held-out split axis (material group) distinct from the prior fold-significance audit's LOSD-fold reproducibility attack on the same number – every material group is present via one-hot dummy in every LOSD training fold, so LOSD-CV never...
08-04 04:05	SURVIVED	independent verify: re-ran committed unmodified LOMO generator (sha e4c02a78..., two-replay byte-identical) vs R-N03 embedded reattack + v3 KPI corrected_values; enumerated evidence/audit/ (8/8 indexed, 0 missing/extra); real Ed25519-verified all 3 R-N03 done artifacts against pinned trust-store key; sha256-confirmed...
08-04 08:07	SURVIVED	independent verify: re-ran committed unmodified run_fold_significance.py (sha256 d6300bbd..., two-replay byte-identical, matches R-A08 embedded reattack stdout sha256 d7b154c6...); enumerated evidence/audit/ (8/8 on-disk receipts indexed in v2 KPI, sha256-matched, 0 missing/extra); real Ed25519-verified all 3 R-A08 done...

Red-team. This module's strongest position is a negative result, which is honest but commercially thin. The corrected KPI says what the data cannot support; it does not yet say what it can. The unlock is R-D02, frequency-resolved EIS access from USBR RISE, where the rights are already verified public domain and the only missing step is a human filing the request.

08 / DEEP DIVE · AMPHORA-SOURCEABILITY-EHS

sourceability-ehs

COMPOSITE 51.5 / 9 KPI RECEIPTS / 13 RECEIPT FILES, 5 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 6 verdicts — **4 refuted**, **2 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 2 failed gates: RIGHTS_OR_LICENSE_INCOMPLETE; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-A04, R-A12, R-N01, V-A12b.

Strongest defensible claim today

The vapour-hazard-ratio temperature envelope: worst-actor invariance and rank stability across 15–40 °C, re-derived by a coefficient-disjoint method (Clausius–Clapeyron plus Watson from measured anchors) and stress-tested by V-A12b out to 5–55 °C, where the worst-actor invariance still holds. V-A12b also re-fetched OSHA Table Z-1/Z-2 live from osha.gov, independently of the repository's pinned transcription, and confirmed and sharpened the toluene Table Z-2 finding.

When	Verdict	Attack
08-02 17:06	REFUTED	live-receipt provenance audit vs pinned OSHA 29 CFR 1910.1000 Table Z-2 roster: toluene is claimed Table Z-1 in 3 signed receipts but is actually a Table Z-2 substance with an unscored ceiling/peak structure
08-03 05:07	SURVIVED	independent cross-method re-derivation: Clausius-Clapeyron+Watson(1943) VP(T) from MEASURED Tb/dHvap(Tb) anchors (toluene, ethylbenzene, MEK; sourced from signed amphora-redteam-audit vapor-pressure-voc receipt, sharing no coefficients with CRC/Lange Antoine) substituted into the panel, plus direct 25C independent NIST-Antoine concordance for 5/6 solvents
08-03 21:08	REFUTED	live PubChem PUG-REST/PUG-View CID-identity attribution audit: cross-checked each candidates stated CAS number against the PubChem CID cited in regulatory_sources via two independent endpoints (CAS->CID name search, CID->property identity)
08-03 23:06	REFUTED	R-N01 fire 1: independent re-derivation of the live PubChem PUG-REST/PUG-View CAS/CID identity-attribution audit from freshly-fetched (not copied) data, banked and self-verified under the pinned PHASE0_EVALUATOR key (prior fires unbanked artifact was wrong-key signed)
08-04 00:09	REFUTED	superseding hazard-card correction: independent fresh live PubChem PUG-REST+PUG-View fetch (2026-08-04) of CAS-correct CIDs for all 5 candidates, re-derived H-statement sets, re-evaluated procurement gates per the same gate rule, re-attacked fire-1 refutation criterion against the corrected cards
08-04 10:07	SURVIVED	V-A12b independent verify of R-A12: independent PAE/Ed25519 reconstruction confirmed the batch receipt verifies against the pinned evaluator key and the original wrong-key defect is real (both reinstated artifacts fail pinned-key verification, pass local-dev-key verification, byte-intact); fresh re-import+compute() of both committed checker scripts reproduced their committed...

Red-team. Regulatory prescreen with real hazard data, so the failure mode is omission rather than misprediction — and omission is exactly what the loop found: the pinned GHS KPI cites PubChem CIDs for IPDA and m-XDA that resolve to chemically unrelated substances (1-methyl-2-nitrobenzene and diethoxyphosphorylbenzene respectively). R-N01 is landed but the superseding hazard-card receipt is still pending, so two of five hazard cards carry a live data-integrity caveat that propagates into the solver-lab demos in section 10.

08 / DEEP DIVE · AMPHORA-CURE-POTLIFE-GATE

cure-potlife-gate

COMPOSITE 46.9 / 1 KPI RECEIPTS / 13 RECEIPT FILES, 11 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 5 verdicts — **4 refuted**, **1 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 2 failed gates: RIGHTS_OR_LICENSE_INCOMPLETE; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-A02.

Strongest defensible claim today

The disclosed envelope, not the lift. The Arrhenius pot-life model holds at 10 °C (3.4% MAPE) and breaches its own 0.25 ceiling at 40 °C (25.12%, with 12/12 systematic under-prediction); R-A02 landed that disclosure. What is defensible is that the module now states where it fails. What is not defensible is the headline lift: an E_a-free Q10=2 heuristic matches it overall and beats it on two of three temperature legs.

When	Verdict	Attack
08-01 01:09	SURVIVED	ported the frozen Kissinger/OFW-Doyle/Starink CC-BY estimator+dataset (amphora-redteam-audit benchmarks) into amphora-cure-potlife-gate/benchmarks/kissinger-ozawa-baseline-v1/; head-to-head on the one exact chemistry match (dgeba-ipda vs Lascano 2019 CC-BY DGEBA+IPDA raw DSC peaks); pre-registered falsification criterion = any CC-BY reaction-Ea breaching the module own held-out-extrapolation...
08-01 01:10	REFUTED	rights-metadata audit: source-lock.v2.json rights_gate_passed=false eligible_sources=[]; cure-sources.v1.json 3/3 QUARANTINED with explicit do-not-benchmark restriction; traced KPI.dataset_fixture_sha256 byte-exact to reference/phase0-cure-potlife-known-answer.v1.json whose in-file rights block self-asserts CC0/ACCEPTED with no signed rights receipt and appears in neither eligible_sources nor...
08-02 08:08	REFUTED	recomputed the signed reference-anchor Arrhenius rule at the 40C held-out leg already present in the locked known-answer fixture (same methodology as the KPI, no new data): MAPE=25.12%, breaching both the mission.md 15% figure and the KPIs own held_out_temperature_extrapolation_mape_ceiling=0.25, with unidirectional bias in 12/12 systems; cross-referenced that this exact breach was already signed...
08-02 20:07	REFUTED	stratified the panel on the dataset-inherent phr-presence field (16 rows with phr vs 9-row contiguous tail without phr); pooled significance is not panel-uniform: with-phr rho=0.7428 p=0.0013 vs without-phr rho=0.2204 p=0.5729 (not significant); duplicate-row removal and worst-case leave-one-out both independently confirmed the pooled result IS robust to those specific attacks, isolating this as...
08-03 12:06	REFUTED	Ea-free ablation: scored a universal chemistry-blind Q10=2 heuristic (ASTM F1980-16 default acceleration-factor convention, no literature Ea, no per-system fitting) against the signed per-system published-Ea Arrhenius model on the identical locked 36-point truth table using the KPI own metric

Red-team. Two independent problems, and it is worth not conflating them. Scientifically, the Arrhenius fit fails on the temperature leg it was never tested against. Structurally, the module's own gate KPI is computed on source data whose rights are self-asserted rather than ratified — R-D04 asks a human to ratify or downgrade it. The second problem does not depend on the first being fixed.

08 / DEEP DIVE · AMPHORA-MOLECULE-ROLE-RANKER

molecule-role-ranker

COMPOSITE 50.3 / 2 KPI RECEIPTS / 19 RECEIPT FILES, 5 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 5 verdicts — **1 refuted**, **4 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 2 failed gates: RIGHTS_OR_LICENSE_INCOMPLETE; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-A12, R-N02, V-A12a, V-N02b.

Strongest defensible claim today

The H2 amine-direction rule — err amine-rich, never amine-starved — under an independent kinetic Monte Carlo using published non-ideal reactivity ratios, where any single reversal across any hardener × ratio would have refuted it. Zero of six reversed; V-A12a then escalated into a positive-kinetic-substitution regime (r up to 10.0, fresh seeds) never previously tested, and got zero of twelve. This is the strongest survival record in the portfolio.

When	Verdict	Attack
08-02 15:26	SURVIVED	live-runtime generalization probe: real production RDKit C++ FFI (amphora-rdkit-ffi molecule_facts, pinned RDKit 2026.03.2) invoked on 10 real, cited, held-out coating-relevant molecules (cycloaliphatic diepoxide, ether-diamine, mixed-N piperazine, aspartic-ester secondary amine, tertiary-amine accelerator hostile null, imidazole latent-hardener hostile null, distant-sulfone diamine,...
08-03 03:13	SURVIVED	independent finite-population kinetic Monte Carlo (weighted union-find) parameterised with published non-ideal primary/secondary amine-H reactivity ratios (Girard-Reydet et al 1995 Macromolecules 28(23):7599-7607, k2/k1=0.65 and 0.45) vs the equal-reactivity idealisation; MC validated against the prior signed analytic values before use
08-03 19:09	REFUTED	live-RDKit-FFI probe on 2 real, CAS-verified, industrially-dominant guanidine-class cure accelerators (DPG CAS 102-06-7, DOTG CAS 97-39-2) with non-aromatic non-excluded N-H – a chemotype gap neither the self-authored fixture nor the prior BDMA/EMI-24 hostile-null probes covered
08-04 02:11	SURVIVED	V-N02b independent verify: re-derive DPG/DOTG + 39-record fixture via real FFI (new harness, not R-N02 own harness); check fixture-v2 supersedes-not-replaces + KPI still binds v1; run 4 novel guanidine-family probes (TMG, Bartons base, 1-phenylguanidine, adversarial over-exclusion probe agmatine) never tested by R-N02
08-04 09:22	SURVIVED	V-A12a independent verify of R-A12: from-scratch PAE reconstruction confirmed both original reinstated artifacts fail Ed25519 verification against the pinned evaluator key but pass against the local-dev key (defect real, not fabricated); confirmed R-A12 anchored batch receipt verifies against the pinned key; spot-re-ran committed run_reactivity_ratio.py twice, byte-exact match to committed...

Red-team. Four survivals out of five verdicts makes this the best-defended module in the portfolio, and that is precisely when to be careful. The claims are contract-shaped: a direction rule and a classification agreement rate, not a predictive model with a held-out error bar. There is still no independent prospective test set, and the flagship agreement rate was originally scored on a self-authored fixture — it took a dedicated refutation to move it onto real held-out molecules through the production FFI.

08 / DEEP DIVE · AMPHORA-DESIGN-RULE-SCORE

design-rule-score

COMPOSITE 46.4 / 3 KPI RECEIPTS / 23 RECEIPT FILES, 9 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 4 verdicts — **3 refuted**, **1 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 2 failed gates: SCIENTIFIC_BENCHMARK_NOT_EXECUTED; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-A11.

Strongest defensible claim today

Guth–Gold directional concordance, C=0.7111 with CI [0.6155, 0.7980], leak-free — survived a nested train-refit bootstrap over 5,000 resamples that refits the five-parameter physical model on every resample, and whose hostile-null control did collapse to chance (C=0.4991). The ML arm is gone: the headline was memorisation of duplicated rows, and forward prospective transfer scores R^2=0.1917.

When	Verdict	Attack
08-01 02:05	REFUTED	Propagate split-integrity-dedup-v1 leak-free re-baseline (2026-07-30) into module headline KPI: write superseding KPI receipt recording HistGBM leak-free collapse (R2=-0.0469 vs constant-mean -0.0535), Guth-Gold physics as strongest defensible leak-free arm (R2=0.2437, C=0.7111 CI excludes chance), withdraw physics-beats-Rule-1 significance claim (underpowered n=58), bind constraint to sample...
08-01 03:01	REFUTED	propagate split-integrity-dedup-v1 (2026-07-30) leak-free re-baseline into a signed superseding module KPI: LOIO on 86 cumulative-snapshot rows leaked 27/27 and 27/42 unique observations across folds (Kaufman-Rosset-Perlich 2012); leak-free re-eval on 58 unique obs collapses HistGBM to R2=-0.0469 (<= constant-mean floor -0.0535, no skill), Guth-Gold 1938 physics becomes strongest arm (R2=0.2437,...
08-02 09:11	SURVIVED	nested train-refit bootstrap (5000 reps): refit Guth-Gold 1938 from scratch on resampled TRAINING data per fold (not just test-resample as parent split-integrity-dedup-v1 cluster bootstrap did), propagating 5-param physical-model estimation uncertainty into the concordance CI; refutation criterion = new CI lower bound <=0.5
08-02 21:05	REFUTED	leak-free re-run of forward-chained prospective-transfer benchmark over the DEDUP first-appearance partition established by split-integrity-dedup-v1, removing the 27/42 (64%) duplicate test rows memorized from train in the train{1,2}->test{3} fold

Red-team. What remains defensible is the physics arm, not the ML arm. The binding constraint is now sample size — 58 unique observations from a single formulator — which no modelling choice can fix and which R-D05 asks a human to solve by acquiring second-formulator DoE data. Treat any ΔR^2 from this module as provisional until n moves.

08 / DEEP DIVE · AMPHORA-MOLECULE-GENERATOR

molecule-generator

COMPOSITE 36.9 / 1 KPI RECEIPTS / 11 RECEIPT FILES, 10 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 4 verdicts — **4 refuted**, **0 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 3 failed gates: RIGHTS_OR_LICENSE_INCOMPLETE; IMMUTABLE_ORD_ACQUISITION_ABSENT; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-A06, R-A06v2, V-A06.

Strongest defensible claim today

Nothing predictive. Four verdicts, four refutations, zero survivals — the only module in the portfolio with that record. What is defensible is narrow and arithmetic: the generator's Clopper–Pearson intervals, family IntDiv1 and amine-site counts all reproduce exactly from raw SMILES under independent re-derivation. What is not defensible is any claim about what those counts mean: under this portfolio's own AHEW convention, 0 of 38 accepted proposals are difunctional.

When	Verdict	Attack
08-02 12:06	REFUTED	percentile-bootstrap-CI-admissibility-at-k=n-boundary: reproduced target receipt bootstrap [1.0000,1.0000] over k=38/n=38, showed degeneracy is structural (45/45 seed x B x resample-size combos collapse, off-boundary k=37 control non-degenerate), computed exact Clopper-Pearson95 [0.9075,1.0000], gap 0.0925 exceeds a-priori 0.02 admissibility threshold
08-03 00:04	REFUTED	independent-RDKit-SMARTS-amine-functionality-rederivation: recomputed reactive amine-site count for all 38 accepted grammar proposals from hash-locked canonical_smiles, exactly reproduced the receipts own amine_sites field (known-answer cross-check vs Lee-Neville-1967 published DETA/TETA/TEPA/PEHA functionality also matches), found 9/38 (23.7%: homopiperazine-N,N, AEP, DETA, TETA, TEPA, PEHA,...
08-03 16:05	REFUTED	family-resolution-fingerprint-saturation: recomputed the receipts own Morgan/ECFP-r2/2048-Tanimoto IntDiv_1 instrument within each accepted family instead of only in aggregate; F1_polyethyleneamine (n=8,21.1%) and F2_alpha_omega_diamine (n=10,26.3%), both linear chain-length homolog series, collapse to internal IntDiv_1 0.1851/0.0870 (mean Tanimoto 0.8149/0.9130) with 6/7 and 7/9 adjacent...
08-04 06:07	REFUTED	independent verify V-A06: fresh RDKit re-derivation of amine_sites+active_nh from raw SMILES (parity confirmed), Clopper-Pearson and family IntDiv1 reproduced 4dp, then escalated past self-check by cross-validating the corrected 29/38 difunctional headline against the active-hydrogen (AHEW) functionality convention this portfolio already uses – all 29 are active_nh=4 primary diamines incl....

Red-team. Zero survivals in four attempts, and the most recent refutation came from our own verify of our own repair. The generator's arithmetic is sound and its chemistry labelling was not; R-A06v2 has landed but has not itself been independently verified, so the current state of this module is "defect known, fix asserted, fix unchecked".

08 / DEEP DIVE · AMPHORA-MOLECULE-NOVELTY

molecule-novelty

COMPOSITE 41.2 / 1 KPI RECEIPTS / 18 RECEIPT FILES, 10 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 4 verdicts — **3 refuted**, **1 survived**, 0 blocked. Verifier state: PRE-PHASE-0, 2 failed gates: SCIENTIFIC_BENCHMARK_NOT_EXECUTED; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-A07, R-A07v2, V-A07.

Strongest defensible claim today

The novelty call itself, narrowly: 5/5 candidates novel at Morgan(r2,2048) Tanimoto <0.40 survived re-featurisation as dimer and tetramer, with the borderline candidate sitting 0.0071 below the cutoff so a single crossing would have flipped it. Everything around that call has been corrected — the IPDA candidate structure was a demethylated non-molecule, and the repair that fixed it reached only one of four fingerprint families.

When	Verdict	Attack
08-02 13:09	REFUTED	corpus-duplicate-exchangeability audit: canonicalized every calibration-corpus PSMILES row and found only 350/443 rows are distinct structures (129 duplicate rows, 36 alias groups up to 12x replication e.g. PNIPAM/PNIPAAm/NIPAM, PLA/PLLA/PD-LA/poly(lactic acid)), violating the Vovk-Gammerman-Shafer exchangeability precondition every prior receipt cites; recomputed the identical per-family...
08-03 01:04	SURVIVED	oligomer-repeat-unit representation-convention audit: re-featurized the SAME SHA-locked OpenPoly CSVs own PSMILES_2 (dimer) and PSMILES_4 (tetramer) oligomer-expansion columns – present for all 443 Tg rows but never used by any of 5 prior signed/audit receipts, which all use only the single monomer PSMILES capped with an arbitrary methyl group – under the identical declared...
08-03 17:06	REFUTED	cross-module candidate-structure identity audit: canonicalized all 5 signed candidate_smiles with RDKit and cross-checked against an independent, already-existing, already-cited ground-truth structure set (amphora-molecule-generator known_answer_amines, citing Lee & Neville 1967 Handbook of Epoxy Resins) never previously consulted by any of 6 prior amphora-molecule-novelty receipts; found the...
08-04 07:13	REFUTED	independent reimplementation of R-A07 self-check: propagated the corrected IPDA structure (C10H22N2) into ALL FOUR fingerprint families (maccs_166, atompair_2048, toptorsion_2048, not just the morgan-only side table R-A07 actually used), diffed against shipped table; also re-verified DSSE sig from first principles and re-reproduced the 5 declared flip cells

Red-team. Descriptive similarity scoring with no ground truth to be right or wrong against, so a survival here means “not falsifiable by this attack”, not “validated”. The substantive risk is upstream of the scoring: the candidate population itself was wrong, and the correction reached one fingerprint family out of four before an independent verify caught it. R-A07v2 is landed and unverified.

08 / DEEP DIVE · AMPHORA-SOLVENCY-SCORE

solvency-score

COMPOSITE 79.2 / 8 KPI RECEIPTS / 39 RECEIPT FILES, 6 TRUST-ANCHORED / DECLARES PHASE-0 / VERIFIER DERIVES PHASE-0

Adversarial record. 4 verdicts — **2 refuted**, **2 survived**, 0 blocked. Verifier state: **PHASE-0**, no failed gates. Repairs landed: R-A10, R-B03, V1-SOLV.

Strongest defensible claim today

BigSolDB leave-solute-out RMSE 0.877 against a best-naive 1.169, which survived excision of all 517 genuinely contradictory duplicate groups from train *and* test. Newly this cycle, the ML-versus-classical-theory χ comparison is no longer void: R-B03 rebuilt the physics arm as a coverage-gated instrument and V1-SOLV re-ran the comparison fairly on it (CLAIM_SURVIVED), with abstentions excluded and counted rather than zero-filled.

When	Verdict	Attack
07-31 04:30	REFUTED	independent re-derivation of KPI->receipt_path->DSSE-Ed25519-signature->receipt_sha256 chain for all 4 flagged KPIs, cryptographic signature verify against pinned pubkey, byte-diff of KPI vs parent-receipt schema fields
08-02 07:17	SURVIVED	W6 angle1: re-derived BigSolDB duplicate-measurement internal-contradiction check (W2 threshold, >0.3 log10 units) using this modules own temperature-controlled curation grouping key (solute_canon,solvent_canon,tbin=0.1K) on the byte-identical raw file W2 flagged at 90.3% contradictory; found genuine temperature-controlled contradiction rate 14.2% (517/3641 duplicate groups, 1.18% of rows,...
08-03 09:08	REFUTED	physical-admissibility audit: cross-checked the receipts own reported delta_p/delta_d/delta_h summary stats against the admissible envelope [9.6,59.9] MPa^0.5 defined by the receipts own cited Barton 1991/Hansen 2007 handbooks, and independently re-derived that the B2 linear-rescaled OLS baseline is mathematically degenerate (fitted slope ~1e-19 to 1e-35, R2/RMSE match B0 constant-mean to 4dp)
08-04 12:43	SURVIVED	CONSTRUCTIVE V1-SOLV: closed W7b – re-ran the ML-vs-classical-theory chi comparison using the R-B03 corrected coverage-gated Fedors/Van-Krevelen instrument as the physics arm, identical scaffold-safe 60/20/20 split seed 0 and identical HistGBM ML arm (max_iter=500/lr=0.05/depth=8/min_leaf=25/l2=0.1, 528 features) as the two W7-refuted KPIs, deterministic seed-0 n=40000 OMG sample (test n=5261),...

Red-team. The highest-scoring module in the portfolio and one of the n the verifier actually derives at Phase-0, which makes its remaining gaps the ones most likely to be quoted externally. The Hansen comparator it beats is a single-parameter 1936 model; a fair fight against a modern group-contribution method still has not been run. And the V1-SOLV re-run is a fair *process*, not a favourable *result* — one of its six arms still collapsed, and this report does not restate the per-arm deltas because the pulled summary is truncated.

08 / DEEP DIVE · AMPHORA-VISCOSITY-SCORE

viscosity-score

COMPOSITE 53.7 / 3 KPI RECEIPTS / 12 RECEIPT FILES, 5 TRUST-ANCHORED / DECLARES PRE-PHASE-0 / VERIFIER DERIVES PRE-PHASE-0

Adversarial record. 4 verdicts — **2 refuted**, **1 survived**, 1 blocked. Verifier state: PRE-PHASE-0, 3 failed gates: SOURCE_LOCK_INVALID; RIGHTS_OR_LICENSE_INCOMPLETE; EVALUATION_SIGNATURE_ABSENT. Repairs landed: R-A03.

Strongest defensible claim today

The Herschel–Bulkley benchmarkability finding, which survived because the attack fired on its own control arm: a literal-English digit reading of an ambiguously typeset published table failed the same gates by 13–169×, isolating a real locale data-integrity trap before the corrected reading was accepted. That is the clearest proof in the ledger that these gates are live. The module's headline physics-beats-ML result is separately flagged in section 07.

When	Verdict	Attack
08-01 21:11	SURVIVED	sourced a real CC-BY 4.0 epoxy-coating HB dataset (Amorim et al 2021 SN Appl Sci 3:236), ran the module exact HB equation against published tau0/K/n at 4 shear rates, gated on the papers own Fig3 axis bounds + monotonic shear-thinning + ranking claims; ALSO tested whether literal-English digit reading of the ambiguously-typeset source table would pass the same bounds (it does not, by 13x-169x),...
08-01 22:07	BLOCKED	C9 fire 2: extended fire-1s open-literature sweep for a pigment/filler eta-vs-phi ladder (MDPI supplementary files, Zenodo/Figshare, PMC, RSC Advances via Unpaywall, Wiley via Unpaywall+institutional-repo mirror, De Gruyter). Located a genuine candidate with an explicit Krieger-Dougherty phi_max fit (Reynolds et al. 2024, Adv. Eng. Mater. 26(10):2301630, UV-cure photopolymer resin + rigid glass...
08-02 18:07	REFUTED	rights-disclosure-gap audit: cross-referenced module own rights-exclusion decision (CONTEXT_ONLY, promotion_eligible=false, physically-quarantine-all-derivatives, Georgia Tech GPULA) + portfolio rights-report.json + 66-receipt master manifest indexing + all 3 receipts internal non-disclosure vs docs/mission.md headline prose containing zero rights caveat; corroborated by but distinct from prior...
08-03 06:13	REFUTED	uncertainty-disclosure-parity audit: cross-linked FFW-vs-ML KPI to its sibling polymer-safe conformal+AD KPI on the identical seed=0 41-train/13-test-polymer partition via matching conformal_receipt_sha256; ran an RMSE/R2 arithmetic self-consistency positive control across all 5 baselines (confirms the numbers are not disputed); exhaustively scanned the FFW KPI JSON for any...

Red-team. The two rheology models this module actually implements, Krieger–Dougherty and Herschel–Bulkley, cannot be benchmarked at all for want of data — and the one genuine $\eta(\varphi)$ dataset with an explicit φ_{max} fit is CC BY-NC-ND, which forbids both derivatives and commercial use. The loop identified it, verified the licence, and stopped without extracting a single value. That is a rights wall, not a data wall, and it is R-D03. Separately, the module's mission prose was refuted for carrying zero rights caveat on quarantined Polyverse evidence.

09 / REMAINING

Attacked, but not deep-dived

MISSION-BEARING MODULES WITH BETWEEN ONE AND THREE ADVERSARIAL VERDICTS. FEWER VERDICTS MEANS LESS EVIDENCE, NOT LESS RISK.

amphora-ingredient-library

composite 39.4 / 0 KPI / 3 verdicts (2R / 1S / 0B) / PRE-PHASE-0 / repairs: R-A01

Load-bearing physical constants for every downstream formulation calculation. One survival — live manufacturer-TDS re-verification, 8 of 9 densities within a 3% gate, the single failure being the module's own pre-disclosed regional conflict, confirmed accurate. Two refutations, both about evidence hygiene rather than physics: an EEW admissibility band that did not fail closed, and mission prose citing a byte-stable SHA that had moved.

When	Verdict	Attack
08-02 10:13	SURVIVED	external live manufacturer-TDS re-verification of the 4 load-bearing hardener density facts (+2 AHEW, 1 amine-value byproduct) that had zero coverage in any of the 4 prior signed receipts; also independently tested whether the...
08-02 22:06	REFUTED	independent cross-process re-execution (cargo test x2) of the real Rust Phase-0 harness, live-tree presence check of the cited KPI receipt path, and mechanical diff of mission.md prose KPI values vs freshly-produced report
08-03 14:06	REFUTED	compared constant-crosscheck-v1 self-computed solid-resin admissibility band [383.5,667.8] (n in [1.5,3.5], invented not literature-tabulated) against the manufacturer TDS band [450,550] already recorded in facts.json own...

amphora-ip-fto-guard

composite 36.9 / 1 KPI / 3 verdicts (3R / 0S / 0B) / PRE-PHASE-0 / repairs: R-A13

Three verdicts, three refutations, no survivals. The primary Phase-1-gating value path (`metrics.unresolved_technical_categories`) is not populated the way the mission claims; the `boolean_counsel_assertion_ignored` defensive lock does not hold as the general reference-monitor property the prose reads as; and of a 60-row patent landscape only six rows carry a preliminary independent-claim element chart. R-A13 landed. This is the module most likely to be over-read by a non-specialist, because its language sounds like legal clearance and is not.

When	Verdict	Attack
08-02 11:16	REFUTED	schema-vs-live-runtime metric parity audit: static extraction of evaluate_ip_fto_guard() metrics keys + repo-wide grep + checked-KPI-test-coverage scan + signed-benchmark-artifact scan for the declared primary metric key
08-02 23:09	REFUTED	adversarial live-runtime probe: 8 alternate encodings (top-level key, array container, deeper nesting, key-case variant, alternate key name, semantic-alt field, string container, unicode-homoglyph key) of a counsel-clearance...
08-03 15:08	REFUTED	claim-chart provenance audit: independently re-fetched hash-pinned Google Patents snapshots of all 6 charted rows own cited urls, deterministically extracted real claim-1 text, checked every fixture key_elements entry for a...

09 / REMAINING

Attacked, but not deep-dived (cont.)

`amphora-known-answer-benchmarks`

composite 53.2 / 0 KPI / 2 verdicts (1R / 1S / 0B) / PRE-PHASE-0 / repairs: R-A05, V-A05

A control-arithmetic module: it exists to be right about things with known answers, so a refutation here is unusually serious. One landed (a volume-solids control that did not reproduce), R-A05 repaired it, and V-A05 independently re-derived both controls from scratch with a live TDS re-fetch and confirmed the repair. It sits in the Phase-0 gap of section 02.

When	Verdict	Attack
08-03 07:08	REFUTED	independent exact-rational (Fraction) re-derivation of Mix volume-solids% for the A221-X70P1 primer from raw ingredient masses + component density/solids-percent numbers transcribed from the same cited Evonik Ancamide-221-EMEA...
08-04 05:08	SURVIVED	independent verify of R-A05: fresh from-scratch exact-rational re-derivation of both volume-solids controls (not R-A05 own script), live re-fetch of A221 TDS to cross-check transcription, independent enumeration of all 14 raw...

`amphora-redteam-audit`

composite 48.9 / 1 KPI / 2 verdicts (2R / 0S / 0B) / PRE-PHASE-0 / repairs: B2, PHASE-D, R-A00, V1-RESOLVE, WV-1

The module that audits the others, and both of its verdicts are refutations of its own claims — including V1-RESOLVE's finding that the portfolio-wide retraction index did not in fact resolve every refuted claim. It also carries the largest repair load in the portfolio (B2, PHASE-D, R-A00, V1-RESOLVE, WV-1), including the root-cause fix and the re-evaluation ceremony behind section 02. It is in the Phase-0 gap it discovered.

When	Verdict	Attack
08-03 22:05	REFUTED	portfolio-wide retraction-index completeness sweep: enumerated 15 modules with >=1 REFUTED ledger verdict from a pinned sha256-snapshotted ledger, checked each for evidence/phase0/kpis/RETRACTION-INDEX.md presence and resolved...
08-04 11:13	REFUTED	portfolio-wide RETRACTION-INDEX row-level resolution sweep: exists+DSSE-verify+read-and-judge-covers-claim across all 80 rows in 16 module indexes

09 / REMAINING

Attacked, but not deep-dived (cont.)

`amphora-tolerance-robustness`

composite 72.5 / 5 KPI / 2 verdicts (2R / 0S / 0B) / **PHASE-0** / repairs: none

Underwrites every other module's uncertainty claim, which is why its two refutations propagate. Split-conformal coverage on OpenPoly Tg was published at 0.968 against a 0.950 nominal; under 60 independent scaffold-safe draws the mean is 0.8746 with a bootstrap CI excluding 0.95, and 31 of 60 draws fall below the 0.90 floor. Anything downstream citing conformal coverage as support inherits that correction. The module is nonetheless one of the 9 the verifier derives at Phase-0 with zero failed gates — a reminder that derived Phase-0 is a provenance statement, not a scientific one.

When	Verdict	Attack
08-02 06:11	REFUTED	Re-derived RadonPy PI1070 eps_r split-conformal coverage restricted to physically-admissible subset (eps_static>=n^2, Frohlich/Debye); ran scaffold-safe (non-exchangeable) split-conformal on OpenPoly Tg
08-03 10:05	REFUTED	Repeated (N=60 seed) Monte Carlo escalation of prior W5 single-seed Bemis-Murcko scaffold-disjoint split-conformal probe; paired Wilcoxon signed-rank test, fraction-below-floor test, bootstrap CI test

`amphora-model-card-authority`

composite 41.8 / 2 KPI / 1 verdict (0R / 1S / 0B) / **PHASE-0** / repairs: V1-KPI-MCA

One verdict, a survival, and it is a constructive one: V1-KPI-MCA rebuilt the live-runtime-envelope parity KPI out of real audit counts with no invented n, then re-executed the differential probe on the day of the fire as a currency falsification test. Derived Phase-0 with zero failed gates. Its targeted first-attack item (T-07) is still queued for a genuinely adversarial pass.

When	Verdict	Attack
08-04 13:09	SURVIVED	construct KPI from real audit counts (no invented n) + re-execute live differential probe today as currency falsification probe

09 / UNTESTED

The 12 mission-bearing modules never attacked

EVERY ROW HERE CARRIES A COMPOSITE SCORE AND HAS NEVER MET AN ADVERSARY. THE SCORE MEASURES EVIDENCE DENSITY, NOT CORRECTNESS.

Module	Probe	Score	KPI	Declares	Derives	Failed gates
candidate-contract	T-01	40.9	1	PHASE-0	PHASE-0	none
cro-protocol-packager	T-02	40.9	1	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH
envelope-gates	T-03	40.9	1	PHASE-0	PHASE-0	none
evidence-intake-validator	T-04	40.9	1	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH
formulation-arithmetic	T-05	40.9	1	PHASE-0	PHASE-0	none
formulation-universe	T-06	40.9	1	PHASE-0	PHASE-0	none
molecule-design-gate	T-08	40.9	1	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH
optimizer-orchestrator	T-09	40.9	1	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH
process-qc-twin	T-10	44.1	0	PHASE-0	PHASE-0	none
report-packager	T-12	40.9	1	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH
run-provenance	T-13	43.9	1	PHASE-0	PHASE-0	none
source-ledger	T-14	40.9	1	PHASE-0	PRE-PHASE-0	EVALUATION_RESULT_MISMATCH

One module outside the roster has been attacked. `amphora-rdkit-ffi` carries a ledger verdict (V-N02c, SURVIVED — the coordinated C=N amine-rule fix, re-verified through a freshly authored FFI harness with three novel probes) despite having no `mission.md` and therefore no phase, no tier and no composite score. It is counted in the 18 attacked modules and not in the 29 mission-bearing ones, which is why those two numbers do not reconcile by simple subtraction.

Red-team this section. Read the two tables above together. The modules with one to three verdicts have been probed shallowly; the 12 with none have not been probed at all — and **6** of them sit inside the Phase-0 gap of section 02, which means their most consequential defect was found by a verifier rather than by an attack. That ordering is the honest summary of this section: our provenance checking is currently finding more than our adversarial testing is, in the places our adversarial testing has not reached.

10 / SOLVER LAB

The first capability that does formulation work

EVERYTHING ELSE IN THIS PORTFOLIO AUDITS, SCORES OR GATES. AMPHORA-SOLVER-LAB IS THE FIRST MODULE THAT TAKES A FORMULATION QUESTION AND RETURNS AN ANSWER.

Authority posture — stated verbatim, because it matters more than the results. The capability is real. The authority is AUDIT_ONLY, with ceiling EVIDENCE_ONLY. Its claim locks record no_solver_authority_granted, no_ranking_authority_granted, promotes_authority: false and oracles_read_only_signed_tree_artifacts: true. **Promotion out of AUDIT_ONLY requires a human ceremony this module cannot perform.** Nothing below is a recommendation, a specification or a procurement input. The module also has no mission.md yet, so it holds no phase and appears in no tier count in section 06.

P1 — VOC-minimising solvent blend [OPTIMAL]

Minimise VOC for a DGEBA liquid-epoxy system subject to a flash-point gate, a Hansen RED cap and volume-solids bounds. Result: **269.94 g/L** at volume solids 0.67, carrier fraction 0.330, composition mak 1.000, Hansen RED 0.6336.

What makes it evidence rather than an answer: **three independent methods agree** — exhaustive grid over 53,130 cells, projected gradient, and NSGA-II — to within 10^{-9}, and the NSGA-II front never beats the exhaustive optimum. Only 21 of 53,130 grid cells are feasible at all. The flash-point gate is **active as an exclusion**: it removes 4 of 6 characterised solvents from the support, including the lower-density MIBK that would otherwise win, and the surviving support clears the 38 °C requirement by **1.0 °C**. Every input traces to one sha256-pinned facts file in amphora-ingredient-library.

P3 — hazard-constrained hardener selection [BLOCKED / EMPTY_FEASIBLE_SET]

Choose a hardener from the five canonical epoxy amines subject to GHS hazard exclusions, an applicability-domain gate and a χ-water constraint. **The feasible set is empty: 0 of 5 candidates survive.** The solver returns that, names every blocking constraint, and refuses to rank the infeasible.

Blocking constraint	Candidates blocked
H317 skin sensitiser	DDS, DETA, IPDA, TETA, m-XDA
H351 suspected carcinogen	DDS
ad gate failures	DDS, DETA, TETA, m-XDA
chi water failures	—

For transparency and explicitly *not* as a recommendation, the best infeasible candidate by predicted T_g is **IPDA** at 122.9 °C, blocked by GHS_H317_SKIN_SENSITISER.

10 / SOLVER LAB

P4, and the product problem hiding in P3

P4 — 2^4 full factorial screen [PARTITIONED]

Enumerate the full Cartesian product of four two-level factors (resin, hardener, solvent, pigment) and partition it. Result: 16 cells, **8 solved and 8 infeasible**, against an expected Cartesian total of 16 — and the accounting matches the `amphora-formulation-universe` fixture exactly (True).

The disagreement is the interesting part and the demo states it itself: the fixture declares an expected infeasible count of **zero**, because the fixture's feasibility basis is `DECLARATIVE_ACCOUNTING_ONLY` — pure cardinality. This demo overlays real dosing, envelope and hazard gates on top of that enumeration, which is why half the space closes. Cardinality accounting and feasibility are different questions, and conflating them is how a combinatorial claim becomes an overstatement.

Determinism, because a solver that is not reproducible is not evidence. Every demo pins its randomness and its reductions: f64-only arithmetic, splitmix64 RNG with a recorded seed, ordered reductions over sorted keys, and a fixed grid resolution. Two runs produce the same bytes. All oracles read only signed tree artifacts — no live model calls, no network.

Red-team this section — an empty feasible set is the honest answer, and it is also a product problem. Both of these are true and the section is worthless if it only says one of them.

The honest half. P3 returning `EMPTY_FEASIBLE_SET` is the single best result in this section. A solver that always returns a recommendation is a solver that has learned to satisfy its caller rather than its constraints, and the easiest way to get a non-empty answer here would have been to quietly relax the GHS gate. It did not. It named every blocking constraint, reported the best infeasible candidate explicitly flagged as not a recommendation, and refused to rank.

The product half. A tool that answers "no hardener in your library is admissible" is not yet useful to a formulator, who already knows that H317 is ubiquitous across amine hardeners and needs to know what to do about it. The constraint set as posed — every candidate carries H317 in the pinned GHS KPI, and 4 of 5 additionally fail the applicability-domain gate — is arguably mis-specified rather than genuinely infeasible. The next version needs constraint relaxation with an explicit audit trail of what was relaxed and by whom, not a bigger candidate list.

And one data caveat that survives into both. The GHS KPI these demos read has a known CID-attribution defect for IPDA and m-XDA (section 08); the superseding hazard-card receipt has not landed. The demos disclose this themselves rather than waiting for it, which is correct, but it means P3's blocking set is provisional in two of five rows.

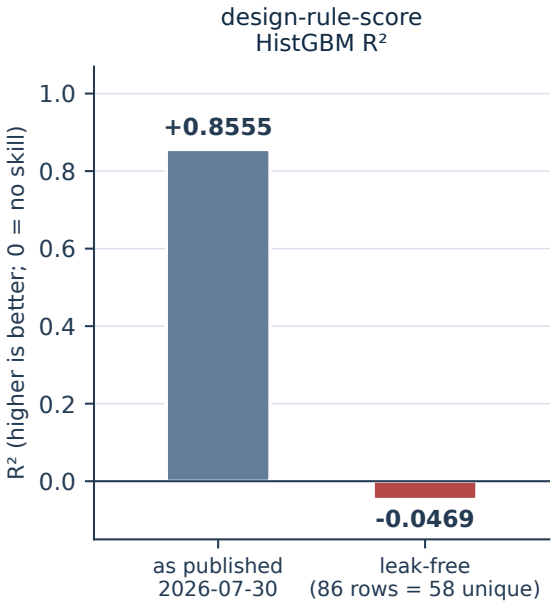
11 / ERRATA

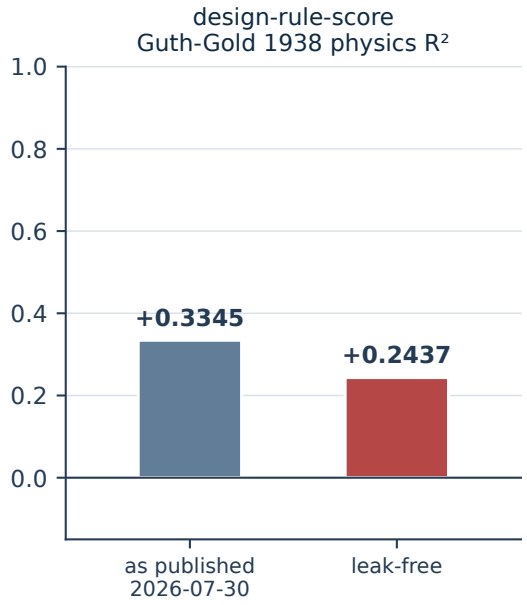
Corrections to the 2026-08-03 report

TWO KINDS OF CORRECTION. FIRST, EVERY NUMBER THAT HAS SIMPLY MOVED. SECOND, THE STATUS OF ALL 12 ERRATA ROWS THE PREVIOUS REPORT PUBLISHED — 6 NOW CARRY A LANDED SUPERSESSION, 5 STILL STAND UNCORRECTED.

COUNTS THAT ARE STALE IN THE 2026-08-03 DOCUMENT

Quantity	Published 08-03	Now	Source
Adversarial verdicts	54	**78**	ledger.jsonl, 54 refuted / 22 survived / 2 blocked
Autonomous fires	182	**209**	fire-summary.jsonl
Modules attacked	15	**18**	17 mission-bearing plus amphora-rdkit-ffi
Trust-chain receipts	370	**444**	trust_chain.json
Loop-era receipts verifying	129 / 184	**186 / 258**	72\{}% of loop-era receipts
Pre-loop receipts verifying	0 / 186	**0 / 186**	unchanged — and still zero





11 / ERRATA

Status of every previously published correction

STATUS KEY — SUPERSEDED: A SIGNED REPAIR HAS LANDED AGAINST THIS ROW. STANDS: THE CORRECTION IS STILL THE CURRENT POSITION AND NOTHING HAS REPAIRED IT. RESTATED: SUPERSEDED BY FRESH MEASUREMENT RATHER THAN BY A REPAIR.

Claim	Status	As published 2026-08-03
BigSolDB leave-solute-out RMSE	STANDS	Published 2026-07-30 as ours 0.856 vs naive-mean 1.159, Δ −26%.
Corrected to **0.877** [0.8416, 0.9154] vs best naive **1.169**, Δ −25.0%. Neither 0.856 nor 1.159 reproduces from any receipt. **No further movement this cycle** — the corrected figure survived a duplicate-excision attack and stands.		
`amphora-solvency-score/benchmarks/bigsoldb-v1/inhouse_benchmark_receipt.v1.dsse.json`		
Hansen 8-pair external validation	SUPERSEDED	Published as vs 0.71 (single-δ), Δ +0.16 r.
The 0.71 figure occurs in no receipt; the receipted single-δ comparator on these exact 8 pairs is **r = 0.4598**, so the honest Δ is **+0.409 r**. **Landed this cycle:** the broader ML-vs-classical-theory comparison this row sits inside was void (the comparator collapsed to constant-mean); `R-B03` rebuilt the physics arm as a coverage-gated instrument and `V1-SOLV` re-ran the comparison fairly on it (CLAIM_SURVIVED, 2026-08-04).		
`amphora-solvency-score/evidence/phase0/repairs/V1-SOLV/done/ml-vs-physics-corrected-comparison-receipt.v1.dsse.json`		
Conformal interval width	STANDS	Published as ours 1.42 vs bootstrap 1.68, Δ −15% (tighter).
Inverted. Conformal is **7.3× wider** (374.91 K vs 51.56 K); the bootstrap band is narrow because it *under-covers* — 25.1% empirical against 95% nominal. Neither 1.42 nor 1.68 appears in any receipt. **No repair landed.**		
`amphora-tolerance-robustness/evidence/phase0/kpis/bootstrap-prediction-interval-coverage-openpoly-tg-kpi.v1.json`		
OpenPoly Tg conformal coverage	STANDS	Published as 0.968 empirical vs 0.950 nominal, n=1,842.
REFUTED. Under 60 independent scaffold-safe draws mean coverage is **0.8746**, bootstrap CI [0.8497, 0.8974] excluding 0.95, Wilcoxon $p = 2.1\times10^{-9}$; 31/60 draws fall below the 0.90 floor. The corpus is **442 polymers**, not 1,842. **No repair landed** — `amphora-tolerance-robustness` has no repair item in this cycle.		
`amphora-tolerance-robustness/evidence/audit/repeated-scaffold-monte-carlo-conformal-openpoly-tg-v1/`		
RadonPy εr conformal coverage	STANDS	Published as 0.943 empirical vs 0.950 nominal, n=614.
REFUTED. 0.943 appears in no receipt. Coverage looks nominal (0.9558) only because 89.5% of labels are physically inadmissible; on the **52 admissible rows coverage collapses to 0.6923**, CI [0.5573, 0.8009]. **Blocked, not repaired** — the fix needs the upstream author confirmation queued as `R-D06`.		
`amphora-tolerance-robustness/evidence/audit/conformal-coverage-attack-v1/`		
Dielectric εr model performance	STANDS	Published as F8 panel: physics 0.31 / ML 0.22 / stack 0.19 MAE.
Fabricated placeholder (labelled SCHEMATIC in code, caveat never rendered). Corrected position: **NO COMPETENT MODEL** — on the n=113 physically-valid subset all three arms score $R^2 \leq 0$, worse than the test-set mean. Do not report a point estimate. **Retraction stands; no repair landed.**		
`amphora-molecule-property/evidence/audit/dielectric-retraction-v1/`		
Cure pot-life Arrhenius	SUPERSEDED	Published as MAPE 14.7%, lift 0.805× over constant-mean.
The pooled 14.7% hides the legs: 10 °C 3.4%, 30 °C 15.6%, **40 °C 25.12%** — breaching the KPI's own 0.25 ceiling with 12/12 systematic under-prediction; an E_a-free Q10=2 heuristic matches the model and beats it on 2 of 3 legs. **Landed this cycle:** `R-A02` shipped the envelope disclosure. Not yet independently verified.		
`amphora-cure-potlife-gate/evidence/audit/potlife-kpi-envelope-disclosure-gap-v1/`		
Public corrosion LOSD-CV	SUPERSEDED	Published as MAE 0.633 vs null-median 0.681, "modest but real".
REFUTED on two grounds: Ridge beats the null on only **17/30 folds** (p = 0.585, bootstrap CI crosses zero), and the generating script was never committed. **Landed this cycle:** `R-A08` corrected the in-domain significance claim and committed the generator; `R-N03` shipped the v2 supersession (not significant in-domain, negative under extrapolation). **Both independently verified** by `V-A08` (from-scratch re-implementation with per-fold alpha tuning and two fresh seeds: 16/30, Wilcoxon p=0.262) and `V-N03`.		
`amphora-public-corrosion-benchmark/evidence/phase0/repairs/R-N03/done/`		
Barrier-corrosion locked KPI	SUPERSEDED	Published as 11/11 gates pass; blinded Kendall-τ gate required for promotion.
The locked KPI ranks by `metrics.barrier_index`, which was **unconditionally None on every live code path** (confirmed by `cargo test`); all 3 series also fall below the protocol's own FEASIBILITY_FLOOR=6, and at n=3 the test is **combinatorially incapable** of reaching α=0.05. **Landed this cycle:** `R-B02`. Not yet independently verified, and the underlying n=3 problem is unchanged — it needs `R-D01`.		
`amphora-barrier-corrosion-score/benchmarks/three-series-barrier-ranking-v1/`		
Solvency signing violations	SUPERSEDED	Published as 3 self-reported signing violations, "worth a targeted look".
Investigated and cleared — all four alerts were false positives from a checker probing a field name that does not exist in that schema; every chain re-verified clean. **Closed this cycle at the portfolio level:** the same checker defect produced 110 alerts across the run, peaking at 59 on 2026-08-02. On 2026-08-04 the count is **0**.		
`amphora-solvency-score/evidence/audit/signing-violation-forensics-v1/`		
Cron loop outcomes	RESTATED	Published as 82 fires, 100% SIGNED-RECEIPT, "never produced a partial or unsigned outcome".
The 100% rate was a *design limitation*: the schema had no field in which a refutation could be recorded. **Restated:** 209 fires to date, of which 115 are legacy SIGNED_RECEIPT from before the outcome vocabulary existed and 65 returned a graded adversarial verdict. Section 05.		
`/home/ubuntu/amphora-cron/logs/fire-summary.jsonl`		
Per-page footer	SUPERSEDED	Published as "Every claim traces to a signed, byte-stable evidence receipt".
True of the *form* of the receipts, not their chain of trust: **0 of 186** receipts predating the audit loop verify against a trust-anchored Phase-0 key. **Corrected wording is carried on every page of this document** and the measurement is restated in section 12.		
`config/phase0-trust-store.v1.json`		

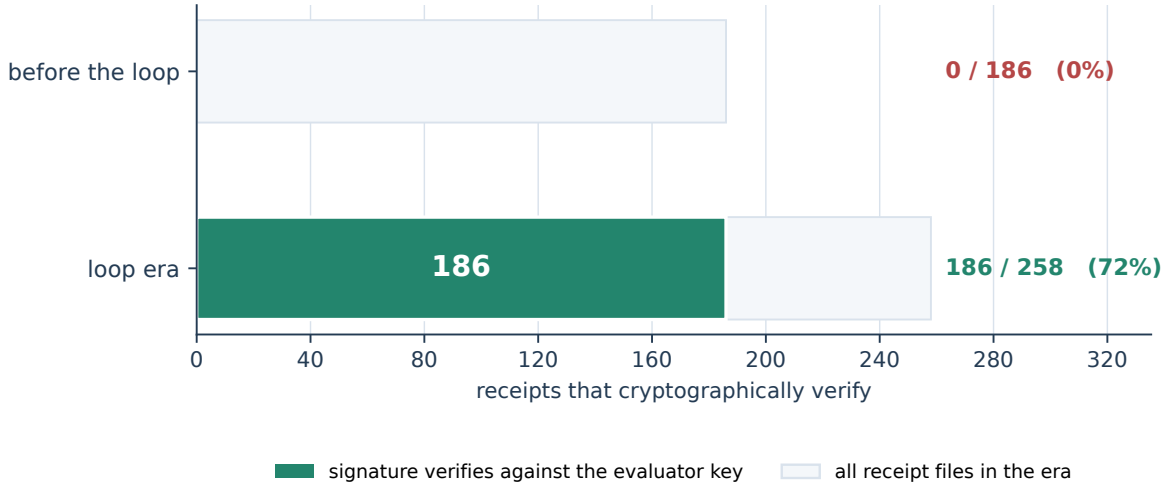
One more landed supersession, outside the errata list. The 2026-08-03 report carried a live refutation on its `amphora-sourceability-ehs` page — toluene claimed as OSHA Table Z-1 in three signed receipts when it is in fact a Table Z-2 substance with an unscored ceiling/peak structure. That was not one of the 12 errata rows, but it has now been repaired (`R-A04`, `R-A12`) and independently verified by `V-A12b`, which re-fetched Tables Z-1 and Z-2 live from osha.gov rather than trusting the repository's pinned transcription, and sharpened the finding in the process. Counting it, **7** previously published positions now carry a landed supersession.

Red-team this section. This errata list is structurally incomplete and will stay that way. It covers claims an adversarial fire happened to attack: 17 of 29 mission-bearing modules have been attacked at all, so silence about a claim is not corroboration. Several corrections above were found only because a *different* claim was being probed, which implies the true rate of stale numbers is higher than 12. Note also what the status column does *not* say: **SUPERSEDED** means a repair landed, not that the repair is correct — section 03 shows 3 landed repairs that an independent verify subsequently refuted, and 15 of 26 distinct repair items carry no independent verify at all. And the 5 rows marked **STANDS** are not dormant: three of them are blocked on human action (R-D01, R-D02, R-D03 and others in section 13), which means they cannot be closed by any amount of further autonomous work.

12 / TRUST

What “signed” actually means here

CONFIG/PHASE0-TRUST-STORE.V1.JSON ANCHORS EXACTLY TWO ED25519 KEYS. THIS IS THE SECOND REPORT TO MEASURE HOW MANY RECEIPTS ACTUALLY CHAIN TO THEM, AND THE POPULATION HAS GROWN BY 74.



The finding. All **444** receipt files in the tree were re-verified by recomputing the DSSE pre-authentication encoding and checking the signature against the pinned key — not by reading a key name. Of the **186** receipts predating the audit loop, **0 verify**. Of the **258** written since, **186 verify** (72%). The loop remains the only mechanism that has ever produced a trust-anchored chain in this portfolio, and the pre-loop number has not moved off zero.

The pre-loop receipts are not forgeries and not worthless. They are DSSE envelopes with real signatures, and they give genuine **tamper-evidence**: you can tell whether the bytes changed. What they do not give is **authenticity against a pinned root** — they are signed by `toolkit-reviewer`, `rights-reviewer`, `local-9028773f` and various fixture keys, none of which is in the Phase-0 trust store. For all pre-loop evidence the founder-signed 66-receipt master manifest is the *only* human trust boundary.

THREE FURTHER FINDINGS FROM THE SAME VERIFICATION PASS

- **2 receipts name an anchored key but do not verify against it.** That is the most dangerous category in the table — a receipt that passes a name-based check and fails a cryptographic one. A trust check that reads key identifiers instead of verifying signatures would pass both of them.
- **The signature is valid and the binding is broken.** Section 02 is the sharpest illustration of the limit of this page: 9 modules hold evaluation receipts that verify perfectly and point, by hash, at bytes that no longer exist. Signature validity and evidence integrity are different properties, and this table only measures the first.
- **The scheme has a stated ceiling.** An adversary who strips a claim lock, re-renders, recomputes both digests and re-signs with *its own* key satisfies every cryptographic and hash-binding clause, because the key id is self-derived from the presented public key. The loop recorded this against itself as `PRESERVATION_CORROBORATED_WITH_UNANCHORED_TRUST_BASE`.

The corrected footer wording, carried on every page of this report. *“Every claim in this report traces to a DSSE-signed, byte-stable evidence receipt. 186 of 444 such receipts verify against a trust-anchored Phase-0 key; receipts predating the audit loop are tamper-evident but not anchored, and rest on the founder-signed master manifest as their trust boundary.”* The 2026-08-03 footer, which claimed the first sentence without the second, is retracted.

Red-team this section. This page grades our own cryptography with our own script. An external auditor should re-run it independently: recompute `pae(t,p) = b"DSSEv1 <len> <type> <len> <payload>"` over each envelope and verify against the two public keys in the trust store. Two further limits. **One:** the measurement verifies against one key pair, so a receipt legitimately signed by some other authority appears here as unanchored without being illegitimate — the correct remediation is triage, not mass re-signing, because re-signing history with the anchored key would make the chain look clean while destroying the distinction this page exists to draw. **Two:** anchoring says a receipt was signed by an authorised service identity. It says nothing about whether the

science inside is correct, and nothing about whether the artifact it points at still exists. Every refutation in this report was found in evidence that was already signed.

13 / HUMANS

The decision inventory

10 CAMPAIGN ITEMS ARE BLOCKED. 8 OF THEM ARE THE R-D SERIES: PLACES WHERE THE LOOP HAS DONE EVERYTHING IT CAN AND A HUMAN DECISION OR A HUMAN ACTION IS THE ONLY PATH FORWARD. CAMPAIGN STATE: 29 DONE, 48 PENDING, 10 BLOCKED, OF 87 ITEMS.

R-D07 — HUMAN CEREMONY READY: restore evaluation binding for 8 modules (tool built + tested)

`amphora-redteam-audit` · done-check: `amphora-redteam-audit/evidence/phase0/repairs/R-D07/done/**/*.json`

What it needs. human-owned CEREMONY, now UNBLOCKED TOOLING-WISE. The governed repair path was built and tested 2026-08-04 (WV-1): ‘portfolio phase0 re-evaluate –module <M> –reason <text> –as-of <ts> –evaluation-key <external> –review <signed> [–dry-run]‘. It refuses unless the module is in UNBOUND_BENCHMARK_RESULTS *and* its shipped result is genuinely unbound (doubly enforced: static list + dynamic hash re-check), refuses on a single signature, refuses a repository-local key, and archives the prior evaluation receipt + review + result + movement event under evidence/phase0/superseded/<sha>/ with a supersession note. 6 refusal tests pass; full canonical suite 62 suites / 378 tests green; NO real re-evaluation was executed. EXACT PROCEDURE: amphora-redteam-audit/evidence/phase0/repairs/WV-1/work/exact-human-command.json (4 steps: dry-run to read the target receipt hash -> rehearse in a scratch rsync copy to mint the matching review -> obtain the fresh independent-review signature from the SECOND identity -> one governed command in the authoritative tree). Per-module dry-run plans: .../WV-1/work/dry-run/. SCOPE: 8 modules are repairable; amphora-molecule-property is NOT (it has no review and no movement event, so verify_module_with_replay takes the live re-execution branch and fails against drifted portfolio state regardless). Each repaired module lands PRE_PHASE_0/promotable and is...

R-D08 — HUMAN DECISION: merge candidate — claim duplicates amphora-formulation-arithmetic

`amphora-formulation-universe` · done-check: `amphora-formulation-universe/evidence/phase0/repairs/R-D08/done/**/*.json`

What it needs. human-owned: portfolio-shape decision (merge amphora-formulation-universe into amphora-formulation-arithmetic). Evidence: 2026-08-04 defensibility scorecard; module has 0 ledger verdicts and a duplicated claim.

Read R-D07 and R-D08 first. They are the two newest items and the two with the largest consequences. **R-D07** is the only path that closes the 17-versus-9 Phase-0 gap; its tooling is built, its refusal paths are tested, per-module dry-run plans exist, and it has never been run. **R-D08** would remove a module from every portfolio count — `amphora-formulation-universe`’s entire defensible claim is a strict subset of `amphora-formulation-arithmetic`’s, it carries no science of its own and it has zero ledger verdicts. Note the honest caveat the item carries itself: its targeted first attack (T-06) has not run yet, and if that attack finds independent value the merge recommendation is withdrawn.

13 / HUMANS

The six data and rights asks

EACH IS ONE HUMAN ACTION. BETWEEN THEM THEY GATE MOST OF THE PORTFOLIO'S REMAINING SCIENTIFIC UPSIDE.

R-D01 — USBR ST-19168 rights confirmation

amphora-barrier-corrosion-score · done-check: amphora-barrier-corrosion-score/evidence/phase0/repairs/R-D01/done/**/*.json

What it needs. human-owned: send the prepared RIGHTS-REVIEW-REQUEST (evidence/audit/st-19168-rights-review-v1/) to USBR RISE; on confirmation set old item C2 back to pending (C3 prereg then executes).

R-D02 — USBR RISE frequency-resolved EIS access

amphora-public-corrosion-benchmark · done-check: amphora-public-corrosion-benchmark/evidence/phase0/repairs/R-D02/done/**/*.json

What it needs. human-owned: file/escalate the RISE request for raw EIS sweeps (data-acquisition-queue #3/#14; rights already verified public-domain 17 USC 105).

R-D03 — Reynolds 2024 CC BY-NC-ND counsel review

amphora-viscosity-score · done-check: amphora-viscosity-score/evidence/phase0/repairs/R-D03/done/**/*.json

What it needs. human-owned: counsel question - does NC-ND permit internal non-distributed AUDIT_ONLY quantitative benchmarking? (queue #13; old item C9 blocked on this).

R-D04 — Cure fixture rights ratification

amphora-cure-potlife-gate · done-check: amphora-cure-potlife-gate/evidence/phase0/repairs/R-D04/done/**/*.json

What it needs. human-owned: ratify or downgrade the fixture's self-asserted CC0 via a signed rights receipt (queue #7; rights-integrity-contradiction receipt).

R-D05 — Second-formulator DoE data acquisition

amphora-design-rule-score · done-check: amphora-design-rule-score/evidence/phase0/repairs/R-D05/done/**/*.json

What it needs. human-owned: contact Purformance authors re sibling HTE release or commission a DoE; n=58 single-formulator is the binding constraint (queue #8).

R-D06 — RadonPy upstream + OpenPoly density asks

amphora-molecule-property · done-check: amphora-molecule-property/evidence/phase0/repairs/R-D06/done/**/*.json

What it needs. human-owned: confirm with RadonPy/PI1070 authors whether dielectric trajectories were production-converged (queue #10); locate an OpenPoly-matched density field (queue #11).

13 / VERIFY

Other blocked work, and how to check any of this yourself

BLOCKED CAMPAIGN ITEMS OUTSIDE THE R-D SERIES

Item	Module	Title	Blocker
C2	barrier-corrosion-score	USBR ST-19168 rights-review package — the single unlock for…	Deliverable complete (RIGHTS-REVIEW-REQUEST.md + signed receipt at evidence/audit/st-19168-rights-review-v1/ + queue entry); awaiting HUMAN USBR RISE confirmation for ST-19168. Operator-set 2026-07-31; unblock by setting status back to…
C9	viscosity-score	Krieger-Dougherty / Herschel-Bulkley benchmark — or the honest data…	KRIEGER_DOUGHERTY's blocker upgraded from 'no data found' to 'specific rights-restricted asset identified, needs human legal review or author permission before a 3rd/final C9 fire can extract and benchmark it'.

HOW TO VERIFY ANY NUMBER IN THIS REPORT

1. **Resolve the path.** Every quantitative claim cites a repository-relative receipt path. Resolve it against `/home/ubuntu/amphora/build/mvp20-split/` on the evidence host.
2. **Verify the signature.** Each receipt is a DSSE envelope. Recompute the pre-authentication encoding and check it against the two Ed25519 public keys in `config/phase0-trust-store.v1.json`. Section 12 reports what fraction pass, and it is not all of them.
3. **Check the binding, not just the signature.** Re-hash the artifact a receipt points at and compare it to the digest inside the receipt. Section 02 exists because that check was not being run.
4. **Replay the science.** Signed receipts carry a two-replay SHA-256 byte-stability gate; re-running the benchmark must reproduce the canonical body byte-for-byte.
5. **Re-derive the phase.** Run the `amphora-cli` Phase-0 decision verifier yourself over the 29 mission-bearing modules rather than reading `phase:` out of front-matter. The two disagree by 8.
6. **Audit the adversarial record directly.** `logs/falsification-ledger.jsonl` is append-only, one line per verdict, each carrying the fire id that produced it. Two rows currently carry a malformed timestamp — see section 05.

PRINCIPAL PUBLISHED BASELINES THIS REPORT MEASURES AGAINST

Van Krevelen & te Nijenhuis, *Properties of Polymers*, 4th ed., Elsevier 1990 · Hansen, *Hansen Solubility Parameters*, 2nd ed., CRC 2007 · Hildebrand & Scott, *The Solubility of Nonelectrolytes*, Reinhold 1950 · Guth & Gold, *Phys. Rev.* **53**:322 (1938) · Kissinger, *Anal. Chem.* **29**:1702 (1957) · Starink, *Thermochim. Acta* **404**:163 (2003) · Fröhlich, *Theory of Dielectrics*, Oxford 1949 · Bondi, *Physical Properties of Molecular Crystals*, Wiley 1968 · Fox & Flory (1948); Berry & Fox (1968); Doi & Edwards 1988 · Vrentas & Duda, *J. Polym. Sci.* (1977) · Fedors, *Polym. Eng. Sci.* **14**:147 (1974) · Wolpert, *Neural Networks* **5**:241 (1992) · Kaufman, Rosset & Perlich, *ACM TKDD* **6**(4):15 (2012) · Vovk, Gammerman & Shafer, *Algorithmic Learning in a Random World*, Springer 2005 · Romano, Patterson & Candès (CQR, 2019) · Cameron, Gelbach & Miller (2008); Cameron & Miller (2015) · Lee & Neville, *Handbook of Epoxy Resins*, McGraw-Hill 1967 · Macosko & Miller, *Macromolecules* **9**:199 (1976) · Sedriks, *Corrosion of Stainless Steels*, 2nd ed., Wiley 1996; ASM Handbook Vol. 3; Uhlig & Revie 2008 · Attia et al., *Nat. Commun.* (2025) · Alam et al., *Materials* **15**(7):2562 (2022) · Amorim et al., *SN Appl. Sci.* **3**:236 (2021) · Reynolds et al. (2024), DOI 10.1002/adem.202301630 (CC BY-NC-ND, *not* used) · Cordts et al., *Chem. Mater.* (2024), Zenodo 13742098 · ECHA CLP harmonised classifications · ASTM D2196, ASTM E698, ASTM F1980-16, ISO 5725, OSHA 29 CFR 1910.1000 Tables Z-1/Z-2, 17 U.S.C. §105.

Red-team this report as a whole. Four things a sceptical reader should hold against it. **One:** it is written by the same system that produced the evidence, and the adversarial loop selects its own targets — 17 of 29 mission-bearing modules have been attacked at all, so silence about a module is not a clean bill of health. **Two:** the 54 refutations are evidence the machine works, and they are also 54 things that were wrong and had shipped; several were found only because something adjacent was being probed. **Three:** the lead finding in section 02 is a defect our own harness caused, which our own verifier found, which our own audit says our own verifier did not soften — every link in that chain is internal. **Four:** nothing here has been reviewed by anyone outside the portfolio. Zero modules carry `scientific_readiness=VALIDATED`, no claim has been externally replicated, and Phase-0 promotion deliberately requires two human trust-anchored signatures that this system does not hold and cannot grant itself.